Team Samarthya

Team ID: 2026-INSPACe-CAN-7USAT-016

IIT Bombay Student Satellite Program

Comprehensive Design Review





Indian Institute of Technology Bombay

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Acknowledgements

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Acronyms

ACK	Acknowledgment
ADCS	Attitude Determination and Control System
ASCII	American Standard Code for Information Interchange
AT	Application transparent
BLDC	Brushless Direct Current
BOM	Bill of Materials
CDH	Command and Data Handling
COMM	Communication Subsystem
CONOPS	Concept of Operations
CSV	Comma-Separated Values
DCS	Descent Control System
DVR	Digital Video Recorder
EKF	Extended Kalman Filter
ELEC	Electrical Subsystem
EPS	Electrical Power System
ESC	Electronic Speed Controller
FPV	First Person View
FSPL	Free Space Path Loss
FSW	Flight Software
GNC	Guidance, Navigation and Control
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GS	Ground Station
HAL	Hardware Abstraction Layer
HDRM	Hold-Down and Release Mechanism
HILS	Hardware-in-the-Loop Simulation
IDE	Integrated Development Environment
IMU	Inertial Measurement Unit
IN-SPACe	Indian National Space Promotion and Authorization Centre
IRCC	Industrial Research and Consultancy Centre
IRNSS	Indian Regional Navigation Satellite System
IITB	Indian Institute of Technology Bombay
IITBSSP	IIT Bombay Student Satellite Program
LOS	Line of Sight
MCS	Mission Control Software
MCU	Microcontroller Unit
MECH	Mechanical Subsystem
NavIC	Navigation with Indian Constellation
NETID	Network Identifier
Ni-MH	Nickel-Metal Hydride
OBC	On-Board Computer
PANID	Personal Area Network Identifier

PCB	Printed Circuit Board
PID	Proportional Integral Derivative
PWM	Pulse Width Modulation
RF	Radio Frequency
RP-SMA	Reverse Polarity SubMiniature Version A
RTOS	Real-Time Operating System
RTC	Real-Time Clock
RW	Reaction Wheel
RX	Receiver
SPI	Serial Peripheral Interface
SRAM	Static Random Access Memory
TX	Transmitter
UI	User Interface
VSWR	Voltage Standing Wave Ratio

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Chapter 1

Team Introduction

The CAN-7USAT team, Samarthya, is part of the IITB Student Satellite Program (IITBSSP), IIT Bombay. IITBSSP was established in 2007 and our first satellite, Pratham was launched in September 2016.
IITBSSP is a 20+ member team currently working on design and development of a 6U CubeSat mission with completely in-house developed modular subsystems. The team operates its own Ground Station.
The team is funded by the Industrial Research and Consultancy Centre (IRCC) at IIT Bombay.
For the CAN-7USAT competition, 7 members were selected from IITBSSP comprising of 2nd, 3rd and 4th-year undergraduate students.
The team follows the Vee model of systems engineering.

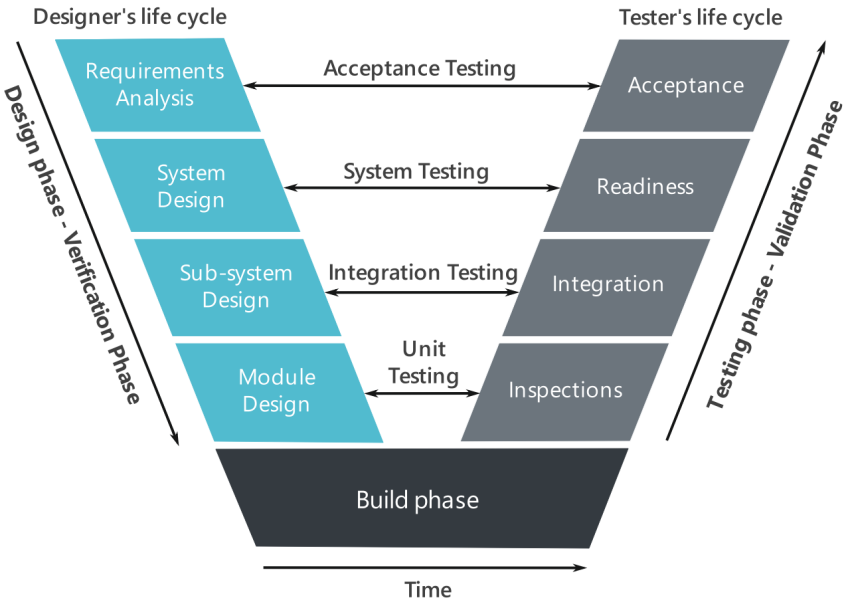


Figure 1.1: Vee Model

Chapter 2

Mission Design

2.1 Mission Objectives

2.1.1 Primary Mission Objective

- Design a CAN-7USAT weighing under 1 kg(+/- 0.050 kg), with dimensions not more than 0.15 m diameter and 0.40 m height.
- Secondary descent control system that shall open at an altitude of 600 m.

2.1.2 Secondary Mission Objective

- Video capture from separation till final touchdown.
- Precision landing control through guided parafoil.

2.2 Mission Architecture

Figure 2.1 illustrates the overall mission architecture, which is divided into a Flight Segment and a Ground Segment. The Flight Segment comprises all onboard subsystems responsible for mission execution, including the Mechanical (MECH), Electrical (ELEC), Attitude Determination and Control System (ADCS), Communication (COMM), Flight Software (FSW), and Descent Control System (DCS) subsystems.

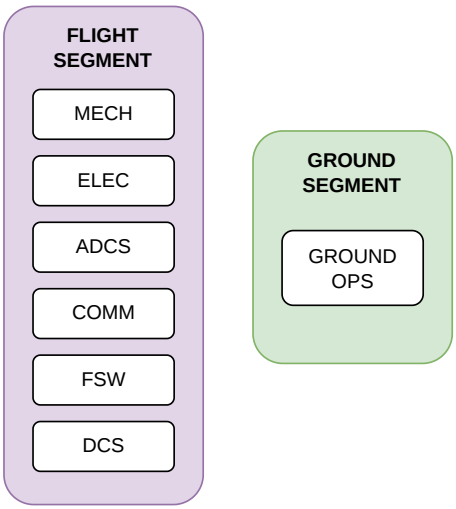


Figure 2.1: Mission Architecture

The Ground Segment consists of the ground station subsystem, which is responsible for ground operations, mission monitoring, telemetry reception, command transmission, re-

covery and post-flight data analysis. Communication between the Flight and Ground Segments is established through the communications subsystem.
The modular architecture facilitates clear subsystem responsibilities, simplifies integration activities, and allows independent development and testing of individual subsystems while maintaining overall mission functionality.

2.3 Requirements

2.3.1 System Requirements

1. Total mass of the CAN-7USAT shall be under 1.000 kg (+/- 0.050 kg).
2. CAN-7USAT shall fit in a cylindrical body of 0.15 m diameter x 0.40 m height.
3. Any sharp edges on the container body shall be avoided.
4. Color of the CAN-7USAT body shall be fluorescent and shall embody the Indian flag.
5. Rocket airframe shall not be used as a part of any CAN-7USAT operation.
6. The CAN-7USAT shall prioritise use of MADE IN INDIA components.
7. The CAN-7USAT shall consist of necessary sensors to provide the following:
    - Position data
    - Altitude
    - Pressure
    - Temperature
    - Power data
    - System status
8. The Global Navigation Satellite System (GNSS) sensor that includes Navigation by Indian Constellation (NavIC) shall be used.
9. The data shall be displayed in near real-time on the ground station User Interface/-Software.
10. CAN-7USAT shall record the data and save it into an onboard SD card in case of telemetry connection loss.
11. The CAN-7USAT shall not transmit telemetry until commanded by the team ground station.
12. The CAN-7USAT shall have a camera which performs video capture upon detection of separation.
13. Total budget shall be less than 1 lakh rupees

2.3.2 Mechanical Requirements

1. All electronics shall be enclosed and shielded from the environment. No electronics can be exposed except for sensors. There must be a structural enclosure.
2. CAN-7USAT structure shall be built to survive 15 Gs of launch acceleration & 30 Gs of shock.

3. Electronic circuit boards must be hard mounted using proper mounts such as stand-offs and screws. High-performance adhesives can also be used.
4. Team number, email address and phone number must be placed on the structure in English, Hindi and the Regional language of the launch state to aid in recovery.
5. Design proper interfacing mechanisms for all components such that the CAN-7USAT is integrable.
6. Provide an easily accessible battery compartment allowing the batteries to be installed or removed in less than a minute.

2.3.3 Electrical Requirements

1. The CAN-7USAT shall have an external power switch with an indicator light or sound for being turned on or off, in order to avoid the dismantling of CAN-7USATs on the launch pad.
2. The CAN-7USAT shall have a battery capacity to support up to 2 hours of wait on the launch pad with additional time for flight operations.
3. The battery source may be alkaline, Ni-Cad, Ni-MH or Lithium ion. Lithium polymer batteries are not allowed. Lithium cells must be manufactured with a metal package similar to 18650 cells.
4. An easily accessible battery compartment must be included allowing batteries to be installed or removed in less than a minute and not require total disassembly of the CAN-7USAT.
5. Spring contacts shall not be used for making electrical connections to batteries. Shock forces can cause momentary disconnects.

2.3.4 Descent Control System Requirements

1. The CAN-7USAT shall contain a total of 2 descent control mechanisms, to be used at different stages while descent.
2. Primary descent control mechanism shall be a parachute while the secondary descent control mechanism shall be decided by the teams.
3. CAN-7USAT shall immediately deploy the first parachute after ejection from the rocket.
4. The first parachute shall be connected to the outer body of the CAN-7USAT and no ejection mechanism shall be attached to it.
5. The descent rate of the primary parachute shall be 20 m/s $\pm$ 5m/s.
6. The second descent control mechanism shall activate at an altitude of 600 m $\pm$ 10 m to further decrease the descent rate of the CAN-7USAT to 1 m/s to 3 m/s.
7. The descent control system shall not use any hazardous chemical-based explosive or pyrotechnic devices.
8. The secondary descent control mechanism shall land CAN-7USAT within 50 m of the designated target point.

2.3.5 Attitude Determination and Control System Requirements

1. CAN-7USAT shall stabilize its attitude in all three axes.
2. CAN-7USAT shall estimate its attitude and provide it to Descent Control System (DCS).

2.3.6 Communication Requirements

1. The CAN-7USAT communications radio shall be the XBEE radio series 1/2/pro or LORA WAN.
2. The LORA WAN shall have their NETID/PANID set to the team number.
3. The XBEE radio shall not use the broadcast mode.
4. The XBEE radio shall not interfere with other XBEE radios.
5. A provision to switch-on and switch-off telemetry of CAN-7USAT to be incorporated through RF Command.
6. Ensure high link margin over a line of sight (LOS) distance of up to 1600m.

2.3.7 Flight Software Requirements

1. The flight software shall maintain and telemeter an indicator of the CAN-7USAT flight software state.
2. In the event of a processor reset during the mission, the flight software shall be able to determine the correct state and mission time/timestamp.
3. Upon powering up, the CAN-7USAT shall collect the required telemetry at a 1 Hz sample rate or more. The telemetry data shall be transmitted with ASCII comma-separated fields followed by a carriage return in the following format:
<TEAMID>,<TIMESTAMPING>,<PACKETCOUNT>,<ALTITUDE>,<PRESSURE>,<TEMP>,<VOLTAGE>,<GNSSTIME>,<GNSSLATITUDE>,<GNSSLONGITUDE>,<GNSSALTITUDE>,<GNSSSATS>,<ACCELEROMETERDATA>,<GYROSPINRATE>,<FLIGHTSOFTWARESTATE>,<OTHERDATA>
4. The CAN-7USAT shall calibrate gyros, barometric altitude, accelerometer after receiving command from the ground station to command the parameters to zero as the CAN-7USAT sits on the launch pad.
5. Telemetry shall include mission time with one second or better resolution.
6. Implement an explicit multi-state tracking mechanism spanning bootup, launch, ascent, and dual descent flight regimes.
7. Maintain execution stability across unexpected CPU restarts using non-volatile storage of last state.
8. Implement closed loop control for stabilization and guided landing.

2.3.8 Ground Operations Requirements

1. An audio beacon shall be installed on CAN-7USAT as a recovery assist. It may be powered after landing or operate continuously. The audio beacon must have a minimum sound pressure level of 92 dB, unobstructed.
2. Team shall develop and use their own ground station. All telemetry shall be displayed in real-time during launch and descent. All telemetry shall be displayed in S.I. units. Teams shall plot data in real-time during flight.
3. The ground station shall command the CAN-7USAT to start transmitting telemetry prior to launch.
4. The ground control station antenna shall be elevated from ground level to ensure adequate coverage and range.
5. Stability of the ground station must be ensured.
6. The ground station shall include one laptop computer with a minimum of two hours of battery operation, XBEE radio/LORA WAN and an antenna mounted on a tripod.
7. The ground station must be portable so the team can be positioned at the ground station operation site along the flight line and if required the team can also move to a different location in case of distant landing location in order to locate the CAN-7USAT.
8. The ground station shall automatically save the telemetry data in CSV format with the filename:

   `Flight_2026-INSPACe-CAN-7USAT-016.csv`

   The CSV file shall contain the mandatory telemetry fields listed in table at 5.4. Additional fields may be appended after the mandatory fields if required by the system design.

2.4 Axis Definition

The axis of CAN-7USAT are defined as in Figure 2.2 In the figure yaw (X) axis is pointing towards the Earth, Roll (Y) axis is into the page and Pitch (Z) completes the right handed triad.

2.5 Concept of Operations (CONOPS)

The mission is divided into six operational phases. Each phase defines its objective, activities, entry and exit conditions, and active elements. The Electrical Subsystem and Flight Software remain operational throughout the mission and are therefore omitted from the active element lists for brevity unless explicitly stated. The Navigation Sensors comprise the GNSS receiver and the barometric sensor, while the ADCS Sensors comprise the IMU and magnetometer. Following deployment from the launch vehicle, the primary parachute is deployed passively as the upper lid separates due to aerodynamic forces. At an altitude of 600 m above ground level, the parafoil is deployed using a burn-wire release mechanism,

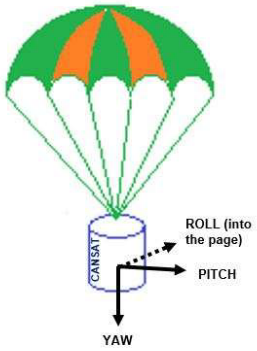


Figure 2.2: CAN-7USAT Axis Definition

after which autonomous guidance and control are initiated. Landing is detected using a combination of altitude monitoring and impact detection, following which the audio beacon is activated to aid recovery.

2.5.1 Phase 1: Ground Handling

Purpose: Operations prior to submission of the CAN-7USAT for launch.

Actions:

- Charge and load battery.
- Boot avionics and flight software.
- Commission communication systems.
- Verify health status of all onboard systems.
- Acquire GNSS lock.
- Verify operation of recovery aids.

Active Elements: All mission elements except parafoil deployment and control elements.

2.5.2 Phase 2: Pre-Flight

Purpose: Launch preparation operations.

Entry Condition: Completion of Ground Handling activities.

Actions:

- Integrate CAN-7USAT into launcher.
- Fold and secure the primary parachute.
- Enable telemetry transmission through telecommand.
- Calibrate sensors.
- Verify flight software readiness
- Attitude estimation

Exit Condition: Launch.
Active Elements:

- COMM.
- Navigational Sensors.
- ADCS Sensors.

2.5.3 Phase 3: Ascent

Purpose: Monitor launch, ascent and deployment detection.
Entry Condition: Launch.
Actions:

- Detect launch using acceleration measurements.
- Transmit telemetry.
- Log mission data.

Exit Condition: Deployment detection of CAN-7USAT from launch vehicle.
Active Elements:

- COMM.
- Navigational Sensors.
- ADCS Sensors.

2.5.4 Phase 4: Primary Descent

Purpose: Provide a stabilized descent following deployment from the launcher.
Entry Condition: Deployment detection of CAN-7USAT from launch vehicle.
Actions:

- Upper lid separates due to aerodynamic forces.
- Deploy the primary parachute.
- Initiate video capture.
- Transmit telemetry.
- Log mission data.
- Activate reaction wheel after approximately 50 m of descent to stabilize yaw.

Exit Condition: Altitude reaches 600 m.
Active Elements:

- COMM.
- Navigational Sensors.
- ADCS Sensors.
- Camera.
- Primary Parachute.
- Reaction Wheel.

2.5.5 Phase 5: Secondary Descent

Purpose: Deploy the parafoil and autonomously navigate towards the target landing location.

Entry Condition: Altitude reaches 600 m.
Actions:

- HDRM losen ups the rods and parachute takes off the bird cage with itself resulting in deployment of parafoil
- Initiate guidance and control algorithms.
- Control the parafoil using the servo actuators.
- Transmit telemetry.
- Log mission data.
- Video capture.

Exit Condition: Landing detected.
Active Elements:

- COMM.
- Navigational Sensors.
- ADCS Sensors.
- Reaction Wheel.
- Camera.
- Parafoil.
- Parafoil Actuators.

2.5.6 Phase 6: Recovery

Purpose: Locate and recover the CAN-7USAT after landing.

Entry Condition: Landing detected.
Actions:

- Activate audio beacon.
- Transmit telemetry.
- Stop logging and save final mission data.

Exit Condition: CAN-7USAT recovered.
Active Elements:

- Audio Beacon.
- COMM.
- GNSS.

2.6 Key Changes from PDR

The following major changes have been made since the Preliminary Design Review (PDR). We have discussed them in detail later in the document.

2.6.1 Flight Software Architecture

The RTOS-based architecture has been replaced with a cooperative super-loop architecture. The GPIO-based mode selection mechanism has also been removed, with separate firmware builds now used for flight and testing.

2.6.2 System Configuration

The mission sequence has been updated such that the primary parachute is deployed immediately after separation, followed by parafoil deployment at an altitude of 600 m using a burn-wire mechanism. Guidance and control commence after parafoil deployment, and landing is detected before activating the audio beacon.

2.7 System Configuration

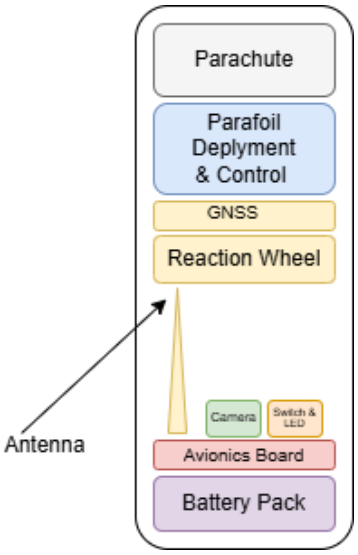


Figure 2.3: System Configuration

Figure 2.3 illustrates the preliminary system configuration. The major design drivers for the selected configuration are listed below:

1. Cylindrical structure with a diameter of 150 mm and overall length of 400 mm, providing adequate volume while remaining within competition constraints.
2. Partially enclosed structural configuration comprising an open upper framework supported by longitudinal rods for the parachute and parafoil deployment systems, and a fully enclosed lower cylindrical section housing the avionics, battery pack, GNSS receiver, reaction wheel assembly, and antenna, balancing mass reduction with structural integrity.

3. Mass distribution designed to position the centre of mass close to the geometric centre of the CAN-7USAT, thereby reducing inertial imbalance and improving controllability.
4. The primary parachute is housed within a dedicated 75 mm open rod-supported upper section, enabling rapid deployment immediately after ejection from the launch vehicle.
5. Environmental sensors such as the barometric sensor and temperature sensor positioned near the CAN-7USAT periphery to improve exposure to ambient conditions.
6. Magnetometer mounted away from electrical systems and actuators to minimise magnetic interference.
7. Easily accessible battery compartment to facilitate rapid installation, replacement, and pre-flight servicing.
8. Vertically aligned dipole antenna positioned along the CAN-7USAT axis to maintain polarization compatibility with the ground station antenna.
9. Major electronic subsystems co-located to simplify harness routing and integration.
10. Reaction wheel incorporated to provide active yaw stabilisation.
11. A dedicated 105 mm rod-supported section houses the parafoil deployment and control system, enabling autonomous transition from primary parachute descent to controlled parafoil flight at an altitude of 600 m.
12. Camera mounted near the outer periphery to maximize field of view during descent, enabling capture of both the CAN-7USAT deployment sequence and ground approach while minimizing obstruction from the parafoil, parachute, and structural elements.
13. GNSS patch antenna mounted on the upper surface of the reaction wheel enclosure and oriented towards zenith.
14. The antenna is positioned within the enclosed lower structure to protect the RF hardware while reserving the upper sections exclusively for parachute and parafoil packaging and deployment.

Chapter 3

Mechanical Subsystem

3.1 Subsystem Requirements

The mechanical subsystem mainly has the task of housing and interfacing all CAN-7USAT components while surviving launch. Its exhaustive list of functions are as follows:-

1. Physically interface all components to a structural member.
2. Survive ground, launch and recovery operations of Samarthya CAN-7USAT.
3. Ensure CAN-7USAT survive mass budget of 1 kg(+/- 0.050 kg).
4. Satisfy CAN-7USAT volume constraint (External envelope of 15cm Diameter and 40cm Height)
5. Design a HDRM (Hold-Down Release Mechanism) to deploy secondary Descent Control System at 600m(+/-10m)

3.2 Structural Design Overview

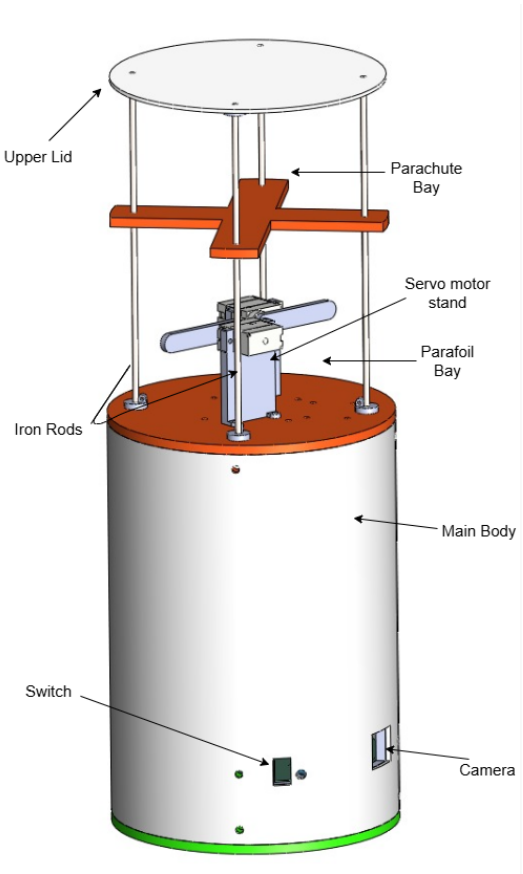


Figure 3.1: CAN-7USAT Structure

The structural is primarily divided into two parts, the Primary Structure which fully encloses all CAN-7USAT avionics and the Descent Control System. Both parts are interfaced and eventually separated via the HDRM.

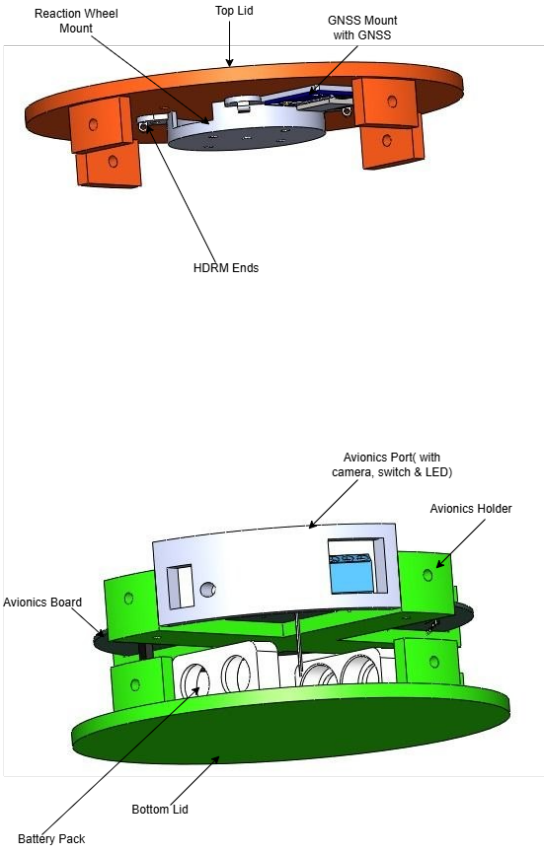


Figure 3.2: CAN-7USAT Structure

The Primary Structure is a completely enclosed cylinder 15cm in diameter with 2.5mm thickness to provide a primary load path during launch. The structure height would be 22cm to reduce mass while providing ample volume for design implementation, along with reducing buckling chances. There is a singular holder to interface avionics board and other components, which also acts as structural load path.

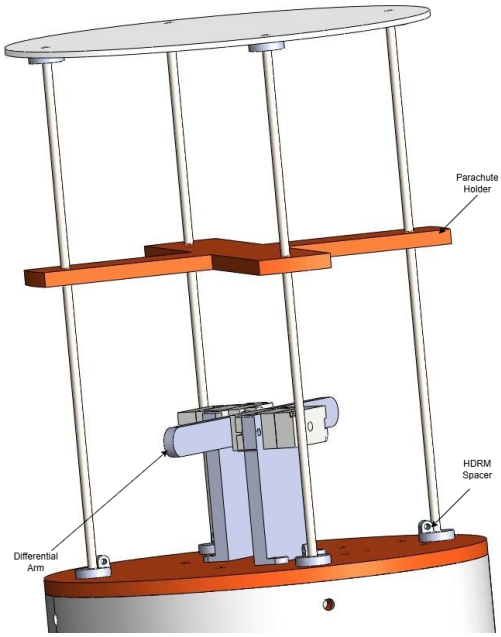


Figure 3.3: CAN-7USAT Structure

The Descent Control System has a birdcage-type structure. It consists of 4 M3 threaded rods acting as primary load path, with another holder interfacing parachute along with being a secondary structural path. The secondary Descent System will be interfaced by the TopLid of Primary Structure. There also exists a Parachute Lid covering the Descent Control System, which comes off post deployment.

3.3 Material Selection & Manufacturing

The lids, holders and main structure of the CAN-7USAT is 3D printed using ABS. The material used in the PCB is FR4.

Component	Material
Battery holder	ABS
Top and Bottom Lid	ABS
Stack Holder	ABS
Primary Structure	ABS
PCBs	FR4

Table 3.1: Materials of Structural Components



Figure 3.4: Manufactured Structure of CAN-7USAT

3.4 Mass Budget

3.4.1 Electrical & Sensor Mass Budget

Components	Mass (in gm)
Battery	150
Servo	26.8
Sensors	1
XBEE	4
Antenna	16
Buzzer	12
BLDC Motor	52.7
ESC	25
Camera	4.8
GNSS	10
DVR	3.5
Total	**305.8**

Table 3.2: Electrical & Sensor Mass Budget

3.4.2 Structural Mass Budget

Components	Mass (in gm)
Battery holder	20
PCB	60
CAN-7USAT Structure	344.5
Reaction Wheel	60
Bolts, nuts, washers, thread lockers and epoxy	30
HDRM	5
Parachute	50
Parafoil	120
Total	**689.5**

Table 3.3: Structural Mass Budget

$$\text{Total Mass Budget} = 305.8 \text{ gm} + 529.5 \text{ gm} = 995.3 \text{ gm}$$

3.5 Mechanical Interfaces

This section describes all the mechanical interfaces between various components and the CAN-7USAT structure.

3.5.1 HDRM

The functions of the HDRM are as follows:

1. Holds Down the main structural elements during launch and deployment till 600m altitude.
2. Deploys the secondary Descent Control System at 600m, effectively separating the two structural elements.

The current implementation is a burn-wire mechanism, where nylon threads whose ends are affixed to the primary structure and the bird-cage structure are burnt via a nichrome resistor.

3.5.2 Mounts & Other interfaces

Special mounts are made for Reaction Wheel, GNSS antenna & Receiver and Parafoil Steering System due to their unique positioning requirements.

Components like the camera, external switch and LED are mounted peripherally, glued to their slots present in the holder.

3.6 Structural Analysis

A simplified geometry has been analysed for launch requirements of 15G acceleration and 30G shock. Since shock profiles are not provided, we have assumed a static load of 30G to get a conservative Margin of Safety.

We have used Static Structural module in Ansys Workbench to derive the equivalent Von-Mises Stresses:-

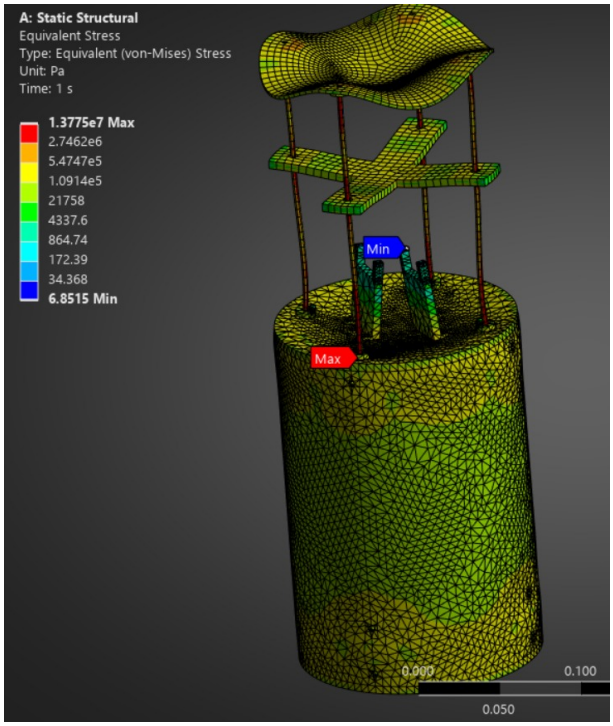


Figure 3.5: Manufactured Structure of CAN-7USAT

Here, peak stress is observed on steel threaded rods at 13.775MPa. Assuming compressive strength of the steel to be 250MPa

$$M.O.S = 250/13.775 - 1 = 17.149 \tag{3.1}$$

For ABS, peak stress observed is 3MPa. Assuming compressive yield strength of ABS to be 45MPa,

$$M.O.S = 45/3 - 1 = 14 \tag{3.2}$$

3.7 Thermal Analysis

3.7.1 Overview

The thermal subsystem aims to maintain operational temperature limits as it undergoes environmental changes during descent from 1 km altitude. A transient thermal analysis is carried out to assess temperature variations during descent.

3.7.2 Component Temperatures

The system components must be kept within their specific minimum and maximum operational and survival temperature ranges to ensure safety during execution.

Component	Min Op (°C)	Max Op (°C)
LG INR18650 Battery	-20	60
Foxeer Predator 5 Nano FPV	-10	50

Table 3.4: Temperature Specifications for Sensitive Components

3.7.3 Thermal Environment

Assuming the CAN-7USAT to be a single lumped body and assuming heat transfer due to radiation being negligible. The operational flight envelope yields a defined descent time map.

$$t_1 = \frac{1000 - 600}{20} = 20 \text{ s}$$
$$t_2 = \frac{600 - 0}{2} = 300 \text{ s}$$
$$T_{\text{total}} = t_1 + t_2 = 320 \text{ s}$$

The variation of altitude $H(t)$ and temperature $T(t)$ of the lumped body across the timeframe are as follows:

$$H(t) = \begin{cases} 1000 - 20t & 0 \leq t < 20 \\ 600 - 2(t - 20) & 20 \leq t < 320 \end{cases}$$

$$T(t) = \begin{cases} 18.5 + 0.02t & 0 \leq t < 20 \\ 18.9 + 0.02(t - 20) & 20 \leq t < 320 \end{cases}$$

The above temperature of the lumped body of CAN-7USAT gives confidence on the decision to not use active thermal control methods. However, there can be local points of concern present near heat generation sources. They shall be resolved via passive thermal design such as thermal straps. A detailed thermal analysis of this shall be done post system design maturity.

Chapter 4

Electrical Subsystem

4.1 Overview of Electrical Subsystem

The electrical subsystem is responsible for providing electrical power, providing computation unit, and making the power budget. The following block design sums up the electrical subsystem.

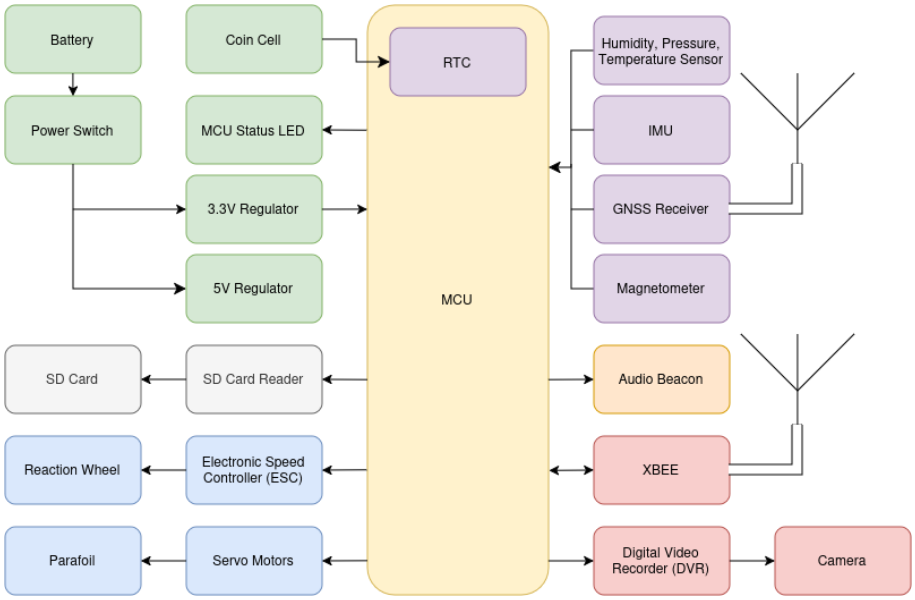


Figure 4.1: Overview of Electrical subsystem

4.2 Battery Charging and Power Distribution

Battery is charged directly from the supply using a battery charging adapter. The voltage from the battery is converted into 3.3V, 5V and 3S unregulated rails to be used by various components and ICs regulators.

4.2.1 Voltage Converters

1. The cells are connected with the voltage converters through a mechanical switch which allow to arm/disarm the electrical subsystem.
2. Battery voltage is read via a resistor divider circuit and fed as an input into the MCU considering the input voltage range of MCU.

- LTC3624EMSE-3.3: 3.3V output
- LTC3624EMSE-25: 5V output

Some specifications of LTC3624 voltage converters are as follows:

- Type: Synchronous Step-Down (Buck) DC-DC Regulator
- Maximum Output Current: 2 A

4.3 Power Management

The power system is responsible for supplying electrical power to all onboard subsystems. It manages power generation and distribution to ensure that each component operates within its specified voltage and current limits and follows a power budget

Table 4.1: Power Consumption and Energy Budget

Component	3.3V	5V	3S Un-regulated	Duty Time	Total Energy (mWh)
MCU: STM32F405RG	79.2 mW			10800s	252.12
Pressure, Temp, Humidity Sensor: AHT20+BME280	0.99 mW			10800s	2.97
IMU: BMI270	2.26 mW			10800s	6.78
Magnetometer: IIS2MDC	0.33 mW			10800s	0.99
XBee-PRO S2C Zigbee	tx: 396 mW, rx: 102.3 mW			10800s	483.12
GNSS Sensor: NEO-6M	600 mW			10800s	1800
Buzzer		120 mW		1800s	75
Camera: Foxeer T Rex Micro FPV		575 mW		320s	51.11
DVR: RunCam Mini FPV DVR		1250 mW		320s	111.11
BLDC Motor A2212 1000KV			6000mW	320s	533.33
Servo Motor MG90S (x2)		1000mW		320s	88.88
Total Load Power	842.84mW	2975mW	6000mW		3405.41

Power consumption for each component is made considering the nominal operating range and input current of components. Most of the components operate at 3.3V/5V, the BLDC motor requires 11.1V for operation. Energy consumption is calculated by multiplying power consumed with its duty time.2

4.4 Microcontroller

To process sensor data, execute flight software algorithm, and control all subsystem operations, a reliable microcontroller is required. The specifications of the microcontroller are as follows:

- Part: STM32F405RG
- Type: 32-bit ARM Cortex-M4 Microcontroller
- CPU Frequency: Up to 168 MHz
- Flash Memory: 1 MB
- SRAM: 192 KB
- Inbuilt RTC module

4.5 Sensors

4.5.1 AHT20+BMP280 Pressure, Temperature and Humidity Sensor

AHT20 and BMP280 sensor module is used to measure ambient pressure, temperature and humidity within the CAN-7USAT.
The specifications of the sensor are as follows:

- Type: Pressure, Temperature and Humidity Sensor
- Pressure Range: 300–1100 hPa
- Humidity Range: 0–100% RH

4.5.2 BMI270 Accelerometer and Gyroscope

BMI270 is used to measure the angular velocity and linear acceleration of the CAN-7USAT during flight. The specifications of the sensor are as follows:

- Type: 6-axis IMU (Accelerometer + Gyroscope)
- Accelerometer Range: $\pm 16g$
- Gyroscope Range: $\pm$2000 dps

4.5.3 IIS2MDC Magnetometer

IIS2MDC is a 3-axis magnetometer which is used to determine the orientation of the CAN-7USAT with respect to the Earth's magnetic field. The specifications of the sensor are as follows:

- Type: 3-axis Magnetometer
- Magnetometer Range: $\pm$50 Gauss

4.5.4 XBEE PRO S2C

XBEE PRO S2C is the wireless communication module used for transmitting telemetry data and receiving commands between the CANSAT and the ground station.

- Frequency Band: 2.4 GHz
- Line of Sight range: 3200 m

4.6 NEO-6M GPS Module

NEO-6M TTL is the GPS module being used to determine the geographical location during flight. This module includes a built-in 25 x 25mm active GPS patch antenna with a UART TTL socket. The specifications of the module are as follows:

- GNSS: GPS
- Horizontal position accuracy: 2.5m

4.7 TPS22917 Switch

The TPS22917 is a load switch used to control the power supply to the buzzer and DVR. The MCU controls these switches.

4.8 Micro SD Card Reader Adapter

- Memory supported: Micro SD Card memory upto 2GB or Micro SDHC up to 32GB

4.9 Camera and DVR

As per the secondary mission objectives, CAN-7USAT will capture a video of descent from separation till touchdown. A camera module, digital video recorder and SD card as storage device will be used to achieve this. The video will later be retrieved from the CAN-7USAT after landing. The components selected are as follows,

- Camera : Foxeer Predator 5 Nano FPV
- Digital Video Recorder : RunCam Mini FPV DVR
- SD Card : SanDisk Ultra microSD

4.10 Buzzer

As per the mission requirements, an audio beacon is activated after landing to aid recovery of the CAN-7USAT. The selected buzzer provides a minimum sound pressure level exceeding the required 92 dB threshold, enabling detection from a considerable distance in the recovery area.
The specifications of the selected buzzer are as follows:

- Model: AI-4228-TWT-R
- Type: Piezoelectric Indicator
- Sound Pressure Level: 99 dBA minimum at 10 cm
- Resonant Frequency: 2.8 kHz

4.11 PCB

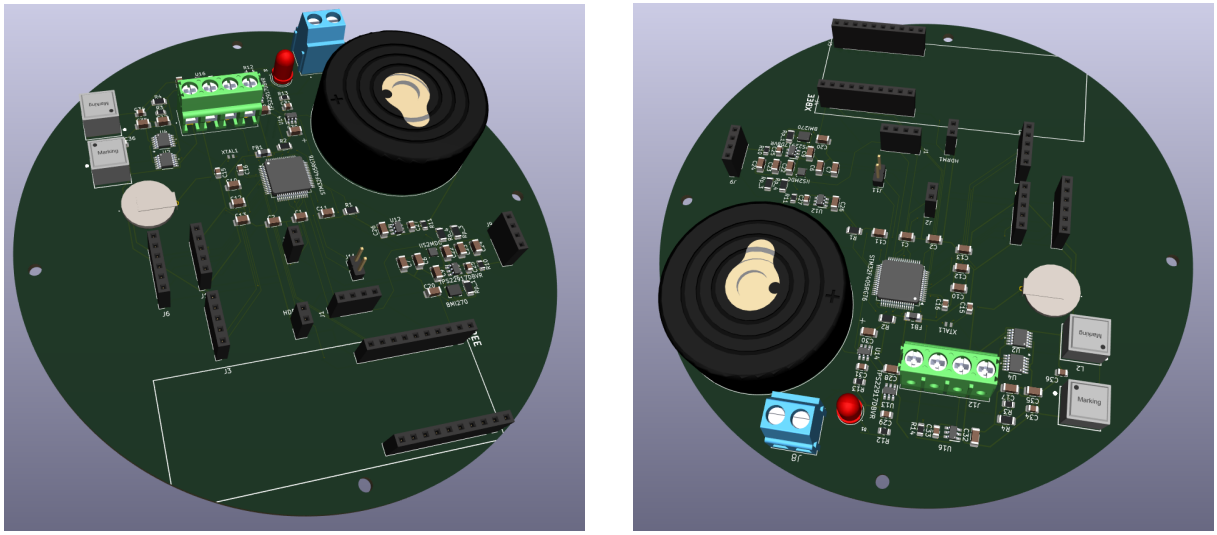


Figure 4.2: PCB 3D view

- PCB is 2-layered and shaped in a circular form with a radius of 15 cm along with 4 mounting holes each of radius 1.5 mm.
- Trace widths for 3.3V, 5V and GND signals are set at 0.5 mm considering a maximum current of 2A and trace widths for the rest signals are set at 0.3 mm considering maximum current flow of 1A.

Table 4.2: Hardware Component Classification

Category	Components
Surface-Mount (SMD)	Microcontroller (`STM32F405RGTX`), IMU (`BMI270`), Magnetometer (`IIS2MDC`), Switch (`TPS229170`), Voltage converters (`LTC3624EMSE-3.3` and `LTC3524EMSE-25`), Passives (resistors, inductors, and capacitors).
Through-Hole (THT)	LED, buzzer, 4-terminal screw terminals (used to connect DVR), 2-terminal screw terminals (used to connect power switch), AHT20+BMP280, XBEE, GNSS sensor, SD card reader module and a 3V coin cell.
Breakout Pins	ST-Link v2 debugger, PWM signals for servos/ESCs, and control signal for parafoil deployment burnwire mechanism.

4.12 Electrical Interfaces

The STM32F405RG interfaces with all onboard peripherals using UART, I^2C, SPI, GPIO, PWM, ADC and SWD interfaces. The electrical interfaces used by the CAN-7USAT

avionics are summarized in Table 4.3.

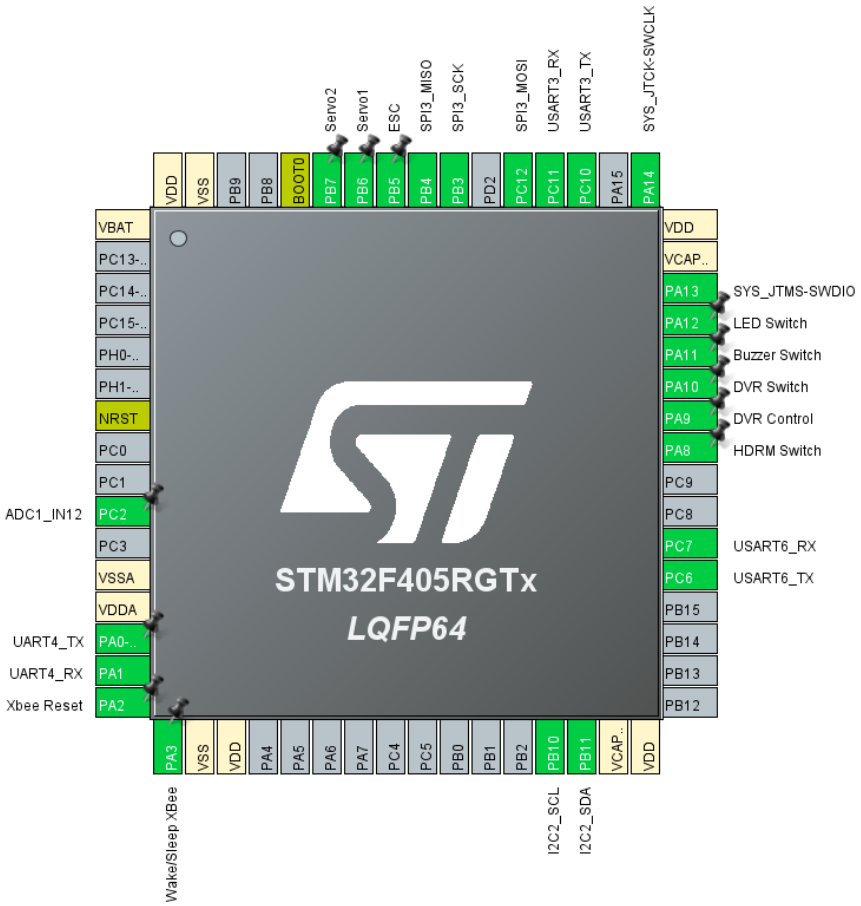


Figure 4.3: STM32CubeIDE I/O configuration

Table 4.3: Electrical Interfaces

Peripheral	**STM32 Interface**
AHT20 + BMP280	I^2C
BMI270 IMU	I^2C
IIS2MDC Magnetometer	I^2C
GNSS Receiver	UART
XBEE Radio	UART
XBEE RESET	GPIO
XBEE ON	GPIO
SD Card Reader	SPI
Battery Voltage Monitor	ADC
LED Switch	GPIO
Buzzer Switch	GPIO
DVR Switch	GPIO
DVR Recording Switch	GPIO
Parafoil Deployment Switch	GPIO
Servo 1	TIM4_CH1 (PWM)
Servo 2	TIM4_CH2 (PWM)
ESC	TIM3_CH2 (PWM)
Debug Interface	SWDIO, SWCLK, NRST
Debug UART	UART

4.13 Testing Progress

The following components of the Electrical Subsystem have undergone preliminary functional testing to verify their individual operation prior to integration with the onboard avionics.

- **AHT20 + BMP280:** The sensor module was intependently interfaced with a Raspberry Pi Pico, and the temperature, humidity, and pressure values were measured.
- **GNSS Receiver:** The GNSS module was intependently interfaced with a Raspberry Pi Pico, and GNSS data was successfully received.
- **Camera and DVR:** The camera interfaced with the DVR was successfully tested to capture video, store it on SD card on the DVR and retrieve it using external SD card reader.
- **SD Card and SD Card Reader:** SD card sitting on the SD card reader interfaced with a Raspberry Pi Pico was tested to read and write data on it.

These tests confirm the correct operation of the individual hardware components. Integration of these peripherals with the STM32-based avionics and system-level validation will be carried out in the next phase of development.

Chapter 5

Communication Subsystem

5.1 Overview of Communication Subsystem

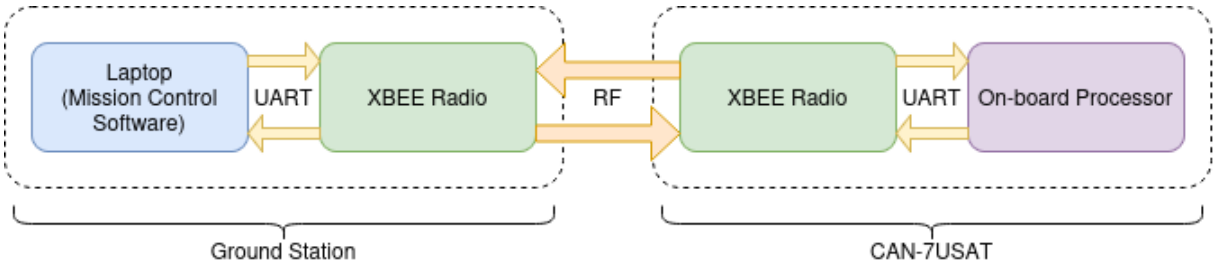


Figure 5.1: Overview of the Communication Subsystem

The communication subsystem is responsible to establish and maintain an RF link between the ground station and the CAN-7USAT.

5.2 Changes Since PDR

- Revised antenna selection as mentioned in section 5.4
- Revised telemetry packet as mentioned in table 5.4

5.3 XBEE Radio

The CAN-7USAT mission uses an **XBEE S2C PRO** module each for the CAN-7USAT and the Ground Station as the RF communication module. Following are the specifications of the same:



Figure 5.2: XBEE Radio

Parameter	Value
Supply Voltage	3.3 V
Frequency Band	2.4 GHz ISM
TX Current	120 mA
RX Current	31 mA
LOS Range	3200 m

Table 5.1: XBEE PRO S2C Specifications

5.3.1 Network Parameters

Both the XBEE operate in **802.15.4** mode. The PANID has been set as instructed by IN-SPACe during frequency allocation stage. Both have been configured to operate with the following network parameters:

Parameter	Value
PANID	0x100F
Network protocol	Zigbee
Transmission mode	Unicast
Encryption	None
Acknowledgement request	Disabled
Onboard XBEE device type	Router
GS XBEE device type	Coordinator
Operating mode	Application Transparent (AT)

Table 5.2: Network Parameters

Frequency Allocation

The chosen XBEE PRO S2C module supports only **channel 11 through 25**. This module cannot work in the channel 26, which had been previously allocated by IN-SPACe during frequency allocation stage.

Acknowledgments (ACKs)

None of the XBEE automatically request or generate ACKs. The onboard MCU is responsible for issuing ACK for received telecommands as mentioned in section 5.6.
Telemetry does not request for ACK.

5.4 Antenna

We will be using 2.4 GHz omnidirectional whip antenna onboard CAN-7USAT for communication with the ground station. The antenna specifications are as mentioned below:



Figure 5.3: Whip Antenna

Parameter	Value
Antenna Type	Whip, tilt
Directionality	Omnidirectional
Frequency Range	2.4 - 2.5 GHz
Impedance	50 Ω
VSWR	1.9:1
Gain	3.2 dBi
Connector Type	RP-SMA
Polarisation	Linear

Table 5.3: Whip Antenna Specifications

5.5 Telemetry

Telemetry transmitted by the CAN-7USAT contains team ID and the real-time parameters as specified in the competition requirements including time, packet count, position, environmental data, battery status, inertial measurement data and flight software state. Payload data including parafoil deployment status and actuator states are also sent in the telemetry. The telemetry is transmitted at **1 Hz** rate by the XBEE radio in **ASCII comma-separated** format. The packet structure is as specified below:

Table 5.4: Telemetry Packet Definition and Data Budget

S.No.	**TM Parameter**	**Function**	**Resolution /Format**	**Bytes**
1	TEAM ID	Team Number	-	27
2	TIME STAMPING	Time since the initial power	0.001 s	3
3	PACKET COUNT	Count of transmitted packets	Integer	2
4	ALTITUDE	Altitude relative to ground	0.1 m	2
5	PRESSURE	Measurement of atmospheric pressure	1 Pa	3

Continued on next page

Table 5.4 – continued from previous page

S.No.	Field	Description	Resolution /Format	Bytes
6	TEMPERATURE	Temperature	0.01 °C	2
7	VOLTAGE	Voltage of the CAN-7USAT power bus	0.01 V	2
8	GNSS TIME	Time generated by the GNSS receiver	0.001 s	4
9	GNSS LATITUDE	Longitude generated by the GNSS receiver	0.0001°	3
10	GNSS LONGITUDE	Longitude generated by the GNSS receiver	0.0001°	3
11	GNSS HEMISPHERE	Hemisphere reported by GNSS receiver	-	2
12	GNSS ALTITUDE	Altitude generated by the GNSS receiver	0.1 m	2
13	GNSS SATELLITES	GNSS satellites connected	Integer	1
14	ACCELEROMETER DATA	Data received from the accelerometer (a_x, a_y, a_z)	1 m/s^2	4
15	GYROSCOPE DATA	Data received from the gyroscopic sensor	0.1 deg/s	6
16	GYRO SPIN RATE	Spin rate of mechanical gyro wrt. CAN-7USAT	Integer	1
17	FLIGHT SOFTWARE STATE	Current flight software state	Integer	1
18	HUMIDITY	Data received from the humidity sensor	1 %	1
19	MAGNETIC FIELD	Data received from the magnetometer	0.01 Gauss	3
20	GNSS SPEED	Speed generated by the GNSS receiver	0.1 m/s	2
21	COURSE	Speed course generated by the GNSS receiver	0.1°	2
22	QUATERNION	Estimated quaternion of CAN-7USAT	0.001	8
23	ANGULAR VELOCITY	Estimated spin rate of CAN-7USAT	0.1°/s	6
24	PARAFOIL DEPLOYMENT STATUS	Deployment status of parafoil	Boolean	1
25	TARGET DISTANCE	Landing point distance	0.1 m	2
26	HEADING ERROR	Heading direction error	0.1°	2
27	LEFT SERVO ROTATION	Angle measurement in left servo	1°	2
28	RIGHT SERVO ROTATION	Angle measurement in right servo	1°	2

Continued on next page

Table 5.4 – continued from previous page

S.No.	Field	Description	Resolution /Format	Bytes
29	DELIMITERS	Comma	-	41

Total packet size: **140 bytes**

5.6 Telecommands

The Ground Station transmits telecommands to the CAN-7USAT through the communication link. Upon successful reception and validation of a telecommand, the onboard MCU executes the requested operation and transmits an acknowledgement (ACK) packet back to the Ground Station. The ACK confirms that the telecommand has been successfully received and processed. If a telecommand is invalid or cannot be executed, a negative acknowledgement (NACK) packet is transmitted to indicate the failure.
The following telecommands are currently supported:

Table 5.5: Supported Telecommands

Command	Description
START_TELEMETRY	Starts periodic transmission of telemetry packets at the configured rate.
STOP_TELEMETRY	Stops telemetry transmission from the onboard system.
CALIBRATE_SENSORS	Initiates the onboard sensor calibration routine before mission operations.
SET_FREQUENCY	Reconfigures the XBEE module to switch to a predefined backup IEEE 802.15.4 communication channel if instructed by the competition organizers or in case of interference on the primary channel.

5.7 Link Budget

- CAN-7USAT EIRP = [Output Power of XBEE (in dBm) - Mismatch losses (in dB) + Antenna gain (in dBi)] dBm

$$\text{CAN-7USAT EIRP} = (18 - 2 + 3.2)\ \text{dBm} = 19.2\ \text{dBm}$$

- Free space path loss (FSPL) = 104.43 dB for a distance of 1600 m for $f = 2475\ MHz$
- Ground station antenna gain = 14 dBi
- Ground station cable losses = 3 dB
- XBEE RX sensitivity = -101 dBm

The RF power loss model equation is given by:

$$\begin{aligned}\text{RX Signal Level} &= \text{CAN-7USAT EIRP} - \text{Free space path loss (FSPL)} \\ &\quad + \text{Ground station antenna gain} - \text{Ground station cable losses} \\ &= (19.2 - 104.4 + 14 - 3)\ \text{dBm} \\ &= -74.2\ \text{dBm}\end{aligned}$$

$$\begin{aligned}\text{Link margin} &= [\text{RX signal level (in dBm) - RX sensitivity (in dBm)}]\ \text{dB} \\ &= [-74.2 - (-101)]\ \text{dB} \\ &= 26.8 dB\end{aligned}$$

Hence, the system will maintain a robustness against noise since there is **+26.8 dB** link margin.

5.8 Testing Progress

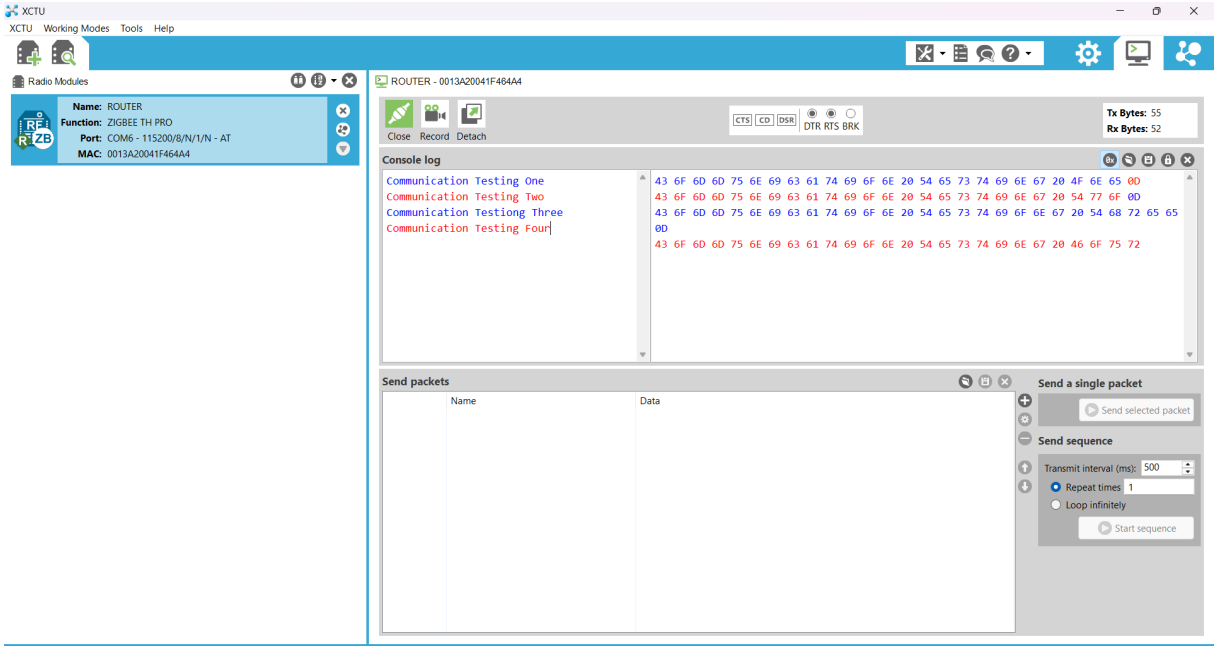


Figure 5.4: XCTU Software

The Communication Subsystem has undergone preliminary functional testing to verify the operation of the XBEE radios.

- Communication between two XBEE S2C PRO modules, configured using the Digi XCTU software, according to the CAN-7USAT network parameters, and each interfaced with a laptop, was successfully established and tested in lab.
- A short-range communication link test was performed using above setup with the onboard and the ground station antenna at the two respective XBEEs, and reliable communication link was established over an approximate distance of 25 m.

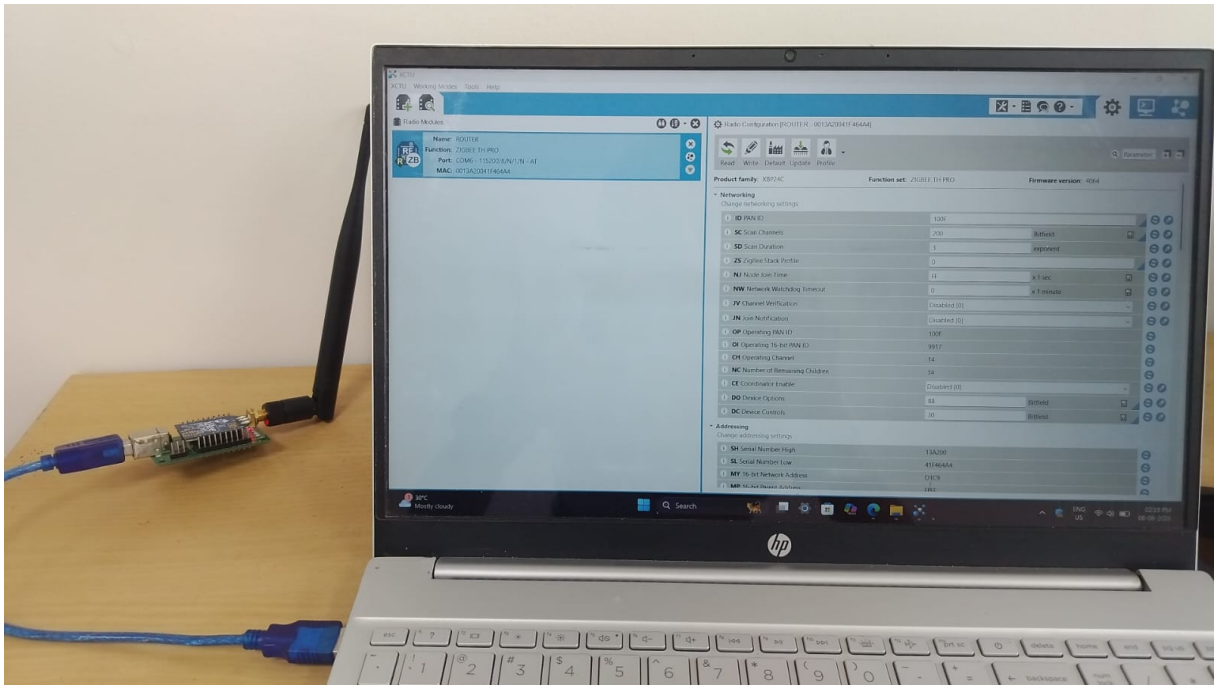


Figure 5.5: XBEE Test Setup

5.9 Future Testing Plan

The following activities are planned to further validate the Communication Subsystem before system integration:

- Validate telemetry transmission and telecommand reception between two XBEEs interfaced with an STM32 MCU and Mission Control Software respectively. Successful execution of telecommands and reception of ACK/NACK responses will be verified.
- Conduct a long-range communication test with the same setup as above over a separation distance of approximately 1.5 km to evaluate communication reliability, packet loss, and received signal strength under field conditions.
- Perform environmental testing of communication system in the various orientations of the integrated CAN-7USAT to verify reliable telemetry transmission at the required update rate and correct handling of telecommands under operational conditions.

The successful completion of these tests will qualify the Communication Subsystem for full system-level integration and mission readiness.

Chapter 6

Attitude Determination & Control Subsystem

6.1 Overview of ADCS

The Attitude Determination and Control Subsystem (ADCS), is responsible for estimating and stabilizing the attitude of the CAN-7USAT throughout its descent. The subsystem enables controlled flight by providing attitude information and generating the required control actions to reduce rotational disturbances and maintain the desired orientation.
Refer 2.3.5 for subsystem requirements.

6.2 Design Overview

The ADCS performs attitude estimation using a Mahony filter that fuses measurements from the IMU and magnetometer to provide real-time estimates of vehicle attitude and angular rates throughout the descent phase.
The estimated angular rates are used by a closed-loop reaction wheel controller to actively stabilize the CAN-7USAT about its yaw axis. The resulting attitude solution is also provided to the Descent Control System (DCS), enabling guidance, navigation and precision landing operations during the final descent phase.

6.3 Reaction Wheel

6.3.1 Momentum Sizing

The required reaction wheel rotation rate and moment of inertia depend on the expected angular momentum acquired by the CAN-7USAT after deployment. In order to get a rough estimate of the angular momentum (L), we assume the CAN7U-SAT as a solid cylinder.

$$I_{body} = \frac{1}{2}MR^2 \approx \frac{1}{2}(1000)(\frac{11}{2})^2 = 15125\ gcm^2$$

Assuming a high angular velocity of $\Omega = 2\pi\ rad/s$ (1 rotation per second) after release,

$$L = I_{body}\Omega \approx 15125 \times 6.28\ rad/s = 94985\ gcm^2/s$$

6.3.2 Reaction Wheel sizing

The reaction wheel should be able to load up the angular momentum calculated above along with any other disturbances. The material selected to make the wheel for this mechanism is acrylic because of the ease in its machining, strength and its sufficient density. Taking $\rho = 1.2\ g/cm^3$. Considering the wheel also as a solid cylinder of radius r and thickness d

In accordance with the mass budget, taking $r = 4\ cm$ and $d = 0.5\ cm$, the mass comes out to be 60 g while ensuring appropriate size . The moment of inertia for the wheel,

$$I_{RW} = \frac{1}{2}(\pi \rho r^2 d) r^2 = 240\ gcm^2$$

6.3.3 Rotor Motor Tradeoff

The reuired angular velocity of the wheel (ω) can be calculated by equating the angular momentum of the wheel and CAN-7USAT,

$$\begin{aligned} I_{RW}\omega &= I_{body}\Omega \\ 240 \times \omega &= 94985 \\ \omega &= 395.8\ rad/s \\ \omega &\approx 3800\ rpm \end{aligned}$$

6.3.4 Motor Selection

The motor chosen is A2212 1000KV BLDC motor rating. It is capable of giving the required angular velocity and torque requirements.

6.3.5 Guidance and Control

A PID based control will be implemented to minimise the yaw rate of CAN-7USAT. The control algorithm will be using the estimated yaw angular rate (Ω_{corr}) from the mahony filter output as described in section 6.4

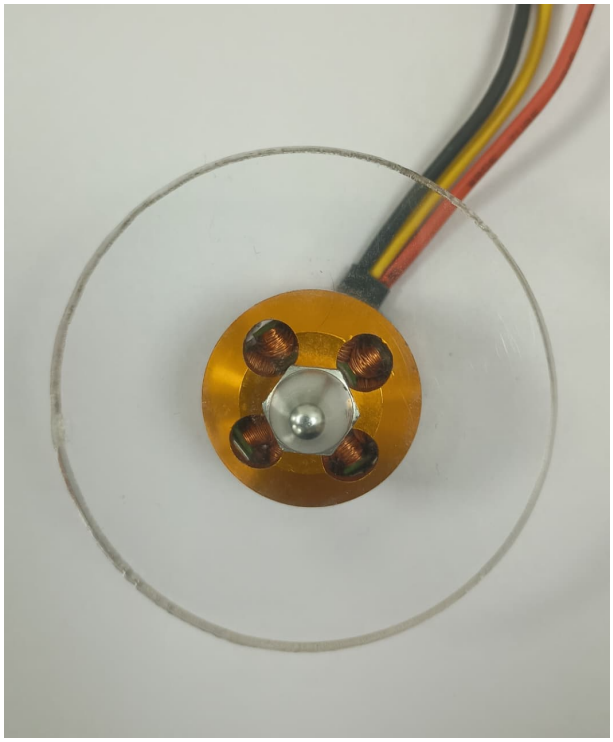


Figure 6.1: Reaction Wheel

6.4 The Mahony Filter

To fuse the 9-DOF IMU data (Accelerometer, Gyroscope, and Magnetometer) into a reliable attitude estimate, the system utilizes an explicit complementary filter formulated on the Special Orthogonal Group $SO(3)$, commonly known as the Mahony Filter.
Traditional navigation systems frequently employ an Extended Kalman Filter (EKF) or Multiplicative Extended Kalman Filter (MEKF). However, these algorithms require continuous calculation of Jacobian matrices and computationally intensive matrix inversions ($O(n^3)$ complexity).
The Mahony filter was selected because it entirely avoids matrix inversion, relying strictly on vector cross-products and a Proportional-Integral (PI) control architecture.

6.4.1 Mathematical Formulation

The Mahony filter operates by integrating the angular rates from the gyroscope to track the attitude, represented as a quaternion $\mathbf{q} = [q_0, q_1, q_2, q_3]^T$. Because gyroscopes inherently drift over time, the filter uses the accelerometer and magnetometer as absolute reference vectors to continuously correct the drift.
First, the algorithm normalizes the measured accelerometer vector ($\mathbf{a}_{meas}$) and magnetometer vector ($\mathbf{m}_{meas}$). It then calculates the expected direction of gravity ($\mathbf{v}_{est}$) and the expected direction of the local magnetic field ($\mathbf{w}_{est}$) based on the current estimated quaternion:

$$\mathbf{v}_{est} = \begin{bmatrix} 2(q_1 q_3 - q_0 q_2) \\ 2(q_0 q_1 + q_2 q_3) \\ q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{bmatrix} \tag{6.1}$$

The filter calculates the angular error ($\mathbf{e}$) between the measured physical reality and the mathematical estimate using the cross product:

$$\mathbf{e} = (\mathbf{a}_{meas} \times \mathbf{v}_{est}) + (\mathbf{m}_{meas} \times \mathbf{w}_{est}) \tag{6.2}$$

This error term is fed into a PI controller to generate a correction vector, which is applied directly to the raw gyroscope measurements ($\mathbf{\Omega}_{\text{meas}}$):

$$\mathbf{\Omega}_{corr} = \mathbf{\Omega}_{meas} + K_p \mathbf{e} + K_i \int \mathbf{e}\, dt \tag{6.3}$$

Where K_p governs the convergence speed of the filter, and K_i acts to eliminate steady-state biases (such as constant gyroscope drift).
Finally, the corrected angular rate $\mathbf{\Omega}_{corr}$ is used to calculate the quaternion derivative, which is integrated over the time step (Δt) to update the system's orientation:

$$\dot{\mathbf{q}} = \frac{1}{2}\mathbf{q} \otimes \begin{bmatrix} 0 \\ \mathbf{\Omega}_{corr} \end{bmatrix} \tag{6.4}$$

$$\mathbf{q}_{t+1} = \mathbf{q}_t + \dot{\mathbf{q}}\Delta t \tag{6.5}$$

6.4.2 Euler Angle Extraction and DCS Integration

While quaternions avoid the singularity problem known as "gimbal lock" during aggressive aerial maneuvers, the DCS navigation logic requires standard Euler angles for its geographic path planning. The absolute heading (Yaw, ψ) is extracted from the normalized quaternion using the following relation:

$$\psi = \arctan 2\left(2(q_1 q_2 + q_0 q_3),\ q_0^2 + q_1^2 - q_2^2 - q_3^2\right) \tag{6.6}$$

This Yaw angle, along with the raw Z-axis gyroscope rate, is passed via a thread-safe double buffer to the DCS task to execute the heading-rate control loop.

6.5 Sensors

6.5.1 BMI270 Accelerometer and Gyroscope

BMI270 is used to measure the angular velocity and linear acceleration of the CAN-7USAT during flight.
The specifications of the sensor are as follows:

- Part: BMI270
- Manufacturer: Bosch Sensortec
- Type: 6-axis IMU (Accelerometer + Gyroscope)
- Accelerometer Range: $\pm 16g$
- Gyroscope Range: $\pm$2000 dps
- Communication Protocol: I2C and SPI
- Operating Voltage: 1.71–3.6 V
- Typical Current Consumption: 685 μA

6.5.2 IIS2MDC Magnetometer

IIS2MDC a 3-axis magnetometer which is used to determine the orientation of the CAN-7USAT with respect to the Earth's magnetic field.
The specifications of the sensor are as follows:

- Part: IIS2MDC
- Manufacturer: STMicroelectronics
- Type: 3-axis Magnetometer
- Magnetometer Range: $\pm$50 Gauss
- Communication Protocol: I2C and SPI
- Operating Voltage: 1.71–3.6 V
- Typical Current Consumption: 100 μA

Chapter 7

Descent Control Subsystem

7.1 Subsystem Overview

The Descent Control Subsystem (DCS) is responsible for safely landing the CAN-7USAT while enabling controlled flight and precision landing. The subsystem employs a two-stage descent architecture consisting of:

1. **Primary Descent System:** A passive parachute deployed immediately after CAN-7USAT ejection from the launch vehicle.
2. **Secondary Descent System:** A GNSS-guided steerable parafoil deployed at an altitude of 600 m above ground level.

The primary parachute reduces the initial descent velocity and stabilizes the vehicle immediately after separation. At 600 m altitude, the secondary descent system is activated, releasing the parafoil and transitioning the vehicle into controlled gliding flight. A dual-servo brake actuation mechanism enables steering of the parafoil, while GNSS-derived position data provides navigation information for precision landing.
The subsystem is designed to satisfy all mission requirements related to descent rate, deployment altitude and targeted landing while remaining within the strict mass and volume constraints imposed by the competition.

7.2 Subsystem Requirements and Design Approach

Requirement	Design Solution
Two independent descent mechanisms	Primary parachute and secondary parafoil
Immediate deployment after rocket ejection	Externally packed parachute attached to top panel
Primary descent velocity below 20 m/s	Sized parachute achieving approximately 20 m/s terminal velocity
Secondary deployment at 600 m ±10 m	Barometric altitude based deployment logic
Final descent velocity between 1–3 m/s	Parafoil sized for approximately 2.5 m/s vertical descent rate
Controlled flight and targeted landing	GNSS based guidance with servo-actuated steering
No pyrotechnic deployment systems	Mechanical HDRM deployment concept
Survive launch and deployment loads	Reinforced attachment points and shock qualified mounting

Table 7.1: Descent Control Subsystem Requirements

7.3 Primary Descent System

7.3.1 Overview

The primary descent system consists of a compact circular parachute packed externally above the CAN-7USAT top panel. This arrangement directly complies with the competition requirement that the primary parachute be attached to the outer body and not require an active ejection mechanism.
The parachute is folded around the top section of the vehicle and restrained using a lightweight retention arrangement. Following launcher ejection, aerodynamic drag extracts and inflates the canopy automatically.

7.3.2 Primary Parachute Sizing

The primary parachute is sized to limit the descent velocity to approximately 20 m/s.
Design Parameters

- CAN7U-SAT Mass: 1 kg
- Target Velocity: 20 m/s
- Air Density: 1.13 kg/m^3
- Drag Coefficient: 0.75

At terminal velocity,

$$\frac{1}{2}\rho V^2 C_D A = mg \tag{7.1}$$

which gives

$$A = \frac{2mg}{\rho C_D V^2} \tag{7.2}$$

Substituting design values,

$$A \approx 0.057 \text{ m}^2 \tag{7.3}$$

The equivalent circular canopy diameter is

$$D = \sqrt{\frac{4A}{\pi}} \tag{7.4}$$

$$D \approx 0.27 \text{ m} \tag{7.5}$$

Selected Design

- Canopy Type: Circular
- Diameter: 90 cm
- Projected Area: 0.636 m^2
- Expected Descent Rate at 600 m: 6 m/s

7.4 Secondary Descent System

7.4.1 Overview

The secondary descent system is a steerable ram-air parafoil designed to provide both descent rate reduction and lateral maneuverability.
Unlike a conventional parachute, the parafoil generates aerodynamic lift, allowing the vehicle to glide toward a predetermined landing location. Steering is achieved through differential braking of the left and right trailing edges.

7.4.2 Deployment Mechanism

The parafoil deployment system employs a non-pyrotechnic Hold Down and Release Mechanism (HDRM) to ensure safe retention and reliable deployment of the parafoil package. The mechanism is described in detail in 3.5.1

7.4.3 Parafoil Canopy Design

The parafoil canopy is designed to provide a controlled final descent with a target vertical descent rate of approximately 3.0 m/s. The selected canopy has a flat span of 1.0 m and a planform area of approximately 0.50 m^2, providing a suitable balance between descent performance, vehicle envelope and controllability.

Canopy Geometry

The preliminary canopy geometry is summarised below.

Parameter	Value
Flat span	1.00 m
Chord	∼0.50 m
Flat planform area	∼0.50 m^2
Flat aspect ratio	∼2.0
Number of cells	7
Number of rib profiles	16
Upper panels	14 (L and R)
Lower panels	14 (L and R)
Suspended mass	1.0 kg
Wing loading	∼2.0 kg/m^2

Table 7.2: Parafoil Canopy Geometry

The canopy consists of seven cells formed using 16 rib profiles, including loaded ribs and half ribs. Cross-ports are incorporated into the ribs to allow span-wise pressure equalisation within the canopy.

The chord is currently estimated from the rib geometry and will be finalised during fabrication. The resulting planform dimensions give an aspect ratio of approximately 2.0.

Predicted Flight Performance

The expected steady-glide performance is estimated for a suspended mass of 1.0 kg. The preliminary aerodynamic model uses an Oswald efficiency factor of 0.85 and an air density of 1.225 kg/m^3.

Parameter	Predicted Value
Lift coefficient, C_L	0.58
Drag coefficient, C_D	0.26
Glide ratio, L/D	2.23
Airspeed	7.1 m/s
Horizontal speed	6.5 m/s
Sink rate	2.9 m/s
Glide range from 600 m	1340 m
Descent time from 600 m	207 s

Table 7.3: Expected Parafoil Flight Performance

The predicted sink rate of approximately 2.9 m/s is within the required final descent velocity range and is close to the target value of 3.0 m/s. The predicted glide ratio of approximately 2.2:1 provides sufficient horizontal range for controlled trajectory correction during descent.

The aerodynamic coefficients and resulting performance values are preliminary estimates. The canopy geometry does not have experimentally characterised aerodynamic

data at this stage, and the actual flight performance may differ from the predicted values. The above parameters are therefore used as the baseline for the preliminary guidance and control design.

Canopy Fabrication

The canopy is fabricated from ripstop fabric using the developed canopy templates. The fabrication geometry includes the upper and lower cell panels, ribs and side ribs, with the required seam allowances incorporated into the templates.

Approximately 2.3 m of ripstop fabric from a 1.4 m wide roll is required for fabrication. The estimated sewn fabric area is approximately 1.4 m^2. For a fabric mass of 38–48 g/m^2, the fabric mass is expected to be approximately 55–68 g, with an estimated finished canopy mass of approximately 100–130 g after accounting for tapes, suspension lines and associated hardware.

Rigging and Control Integration

The suspension-line, bridle and brake-line arrangements are designed separately to integrate the canopy with the Descent Control System. The suspension lines distribute the canopy loads to the payload, while differential brake-line actuation provides directional control.

The brake lines are actuated using servo mechanisms mounted within the CAN-7USAT. Differential actuation of the left and right brake lines produces a change in canopy heading, allowing the Control module to regulate the parafoil heading towards the desired heading generated by the Guidance module.

The final brake-line geometry and actuator travel are selected to provide the required control authority while remaining within the available servo range.

7.5 Guidance and Navigation

7.5.1 Guidance Architecture

The Guidance and Navigation subsystem autonomously guides the CAN-7USAT from the parafoil deployment altitude of 600m to the designated landing location. Following parafoil deployment and canopy stabilization, the subsystem continuously estimates the vehicle state, generates a feasible trajectory, and computes steering commands for autonomous precision landing.

Unlike a conventional parachute, a ram-air parafoil generates aerodynamic lift, enabling controlled gliding flight. The guidance system therefore accounts for the vehicle's turning constraints, glide characteristics and environmental disturbances while navigating towards the target.

The guidance architecture consists of three primary modules:

1. **Navigation:** Estimates the vehicle position, altitude, heading and ground velocity using the GNSS receiver, barometer and Attitude Determination and Control Subsystem (ADCS).
2. **Guidance:** Generates a feasible trajectory to the landing target, tracks it using a Line-of-Sight (LOS) guidance law and performs trajectory replanning when the deviation exceeds a predefined threshold.

3. **Control:** Converts the desired heading into differential brake commands for the parafoil servos to achieve trajectory tracking.

Since the deployment position and heading are unknown prior to launch, trajectory generation is initiated only after parafoil stabilization using the estimated vehicle state. During descent, the Guidance and Navigation subsystem operates as a closed-loop system, continuously updating the reference trajectory based on the latest navigation solution.

Figure 7.1 illustrates the overall Guidance, Navigation and Control architecture adopted for the Descent Control Subsystem.

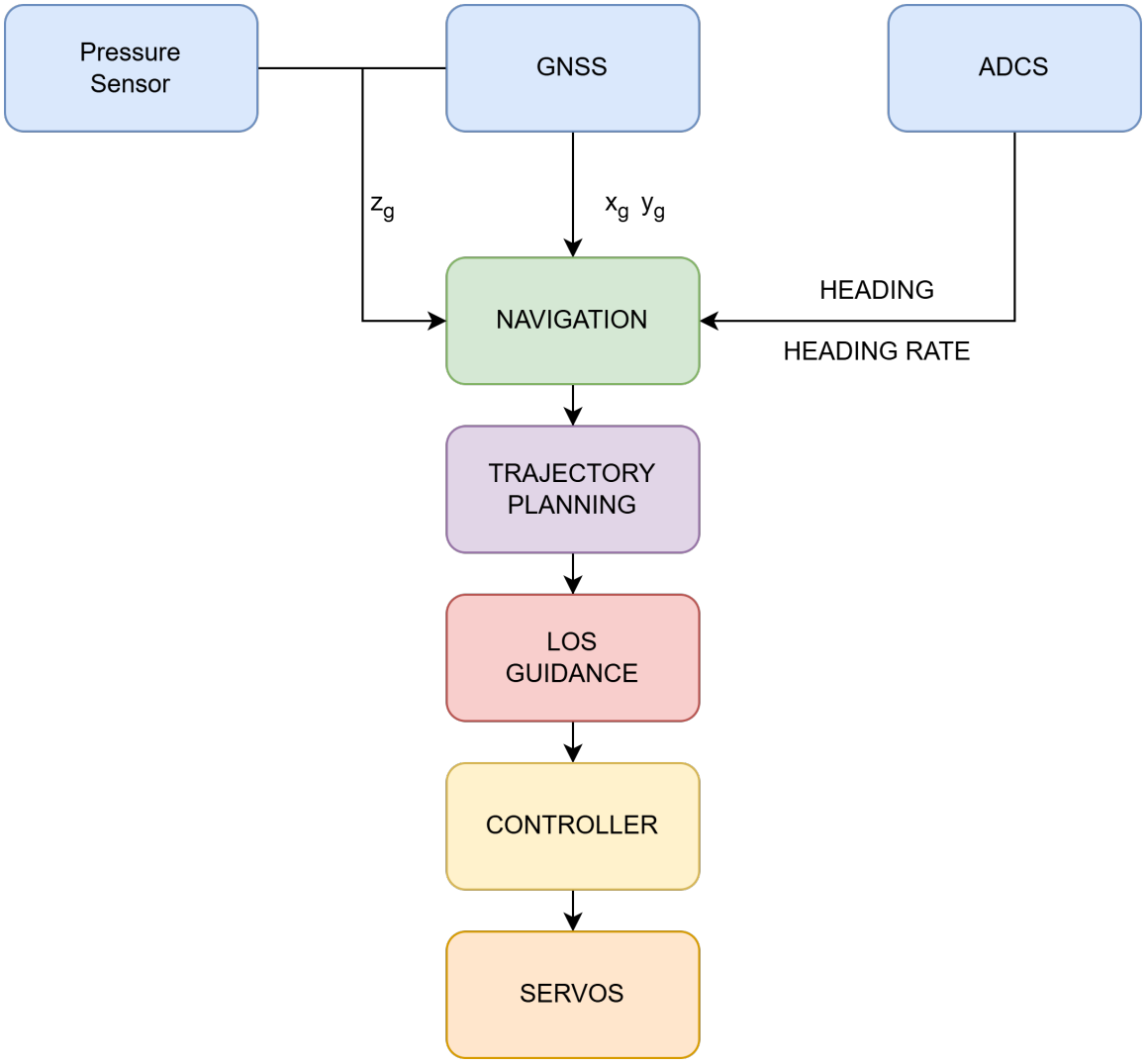


Figure 7.1: Overall Guidance, Navigation and Control Architecture

7.5.2 Coordinate Frames

The Guidance and Navigation subsystem employs multiple coordinate frames to describe the vehicle motion and formulate the guidance algorithm. The adopted coordinate systems are consistent with the reference guidance framework while maintaining compatibility with the body-axis convention established in the Preliminary Design Review (PDR). Figure 7.2 illustrates the coordinate frames considered in the parafoil model.

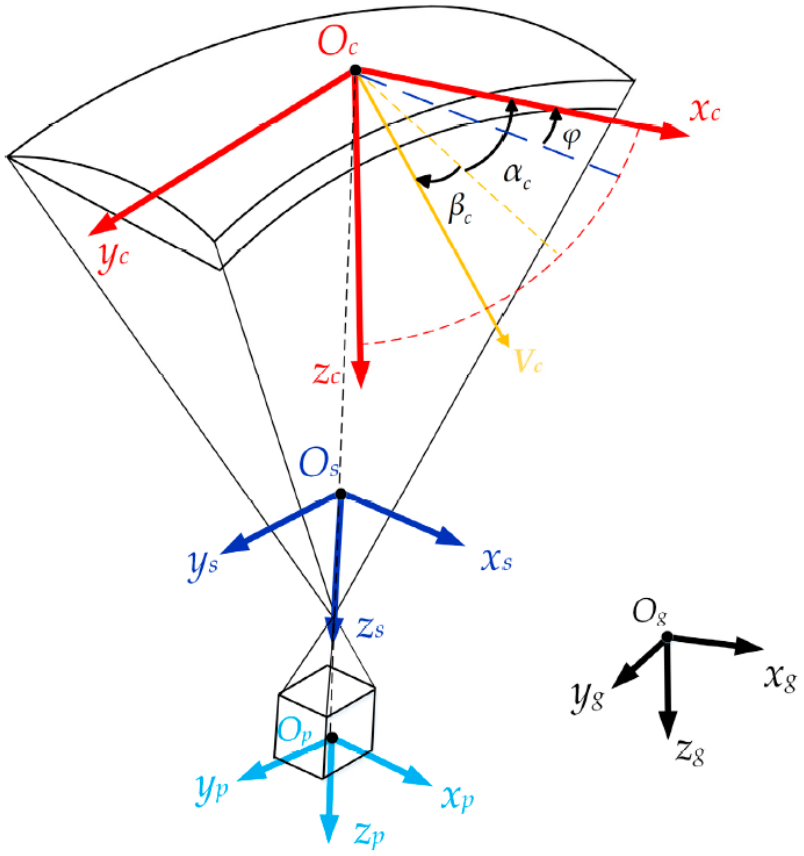


Figure 7.2: Coordinate frames used for the parafoil guidance model.

Ground Frame

A local North-East-Down (NED) frame is adopted as the inertial reference frame for navigation and trajectory planning. Vehicle position, landing target and reference trajectory are all represented in this frame.

Payload Frame

The payload frame is rigidly attached to the CAN-7USAT and follows the axis convention established in the PDR, with the Euler angles roll (ϕ), pitch (θ) and yaw (ψ) describing the vehicle attitude.

Canopy Velocity Frame

The canopy velocity frame is aligned with the instantaneous canopy airspeed. It differs from the canopy frame by the angle of attack (α) and sideslip angle (β). Consistent with the reference guidance model, coordinated flight is assumed throughout the descent, giving

$$\beta = 0.$$

Under this assumption, the velocity vector remains in the longitudinal plane of the canopy, simplifying the guidance formulation.

7.5.3 Modelling Assumptions

The guidance algorithm employs a simplified kinematic model of the parafoil for trajectory planning and control. A complete six-degree-of-freedom aerodynamic model is not adopted at this stage due to the absence of experimentally identified aerodynamic coefficients. The following assumptions are therefore made throughout the guidance and control design.

Zero Sideslip

Consistent with the reference guidance framework, the parafoil is assumed to operate under coordinated flight conditions with negligible sideslip,

This allows the heading angle to adequately represent the direction of motion for trajectory planning.

Constant Glide Performance

The parafoil is assumed to operate near its trimmed flight condition, with approximately constant glide ratio and trim airspeed throughout the controlled descent, except during steering manoeuvres and the terminal flare.

Flat-Earth Approximation

Since the operational altitude and flight range are limited, the Earth is assumed locally flat. Navigation and trajectory planning are therefore performed in a local North-East-Down (NED) frame.

Slowly Varying Wind

Atmospheric wind is assumed to vary slowly relative to the guidance update rate. Consequently, the estimated wind vector is treated as constant over each guidance interval and updated periodically during flight.

Negligible Pendulum Dynamics

The relative motion between the parafoil canopy and payload is neglected for trajectory planning. Residual oscillations are treated as disturbances and compensated by the heading controller.

Simplified Vehicle Model

Trajectory planning is based on a kinematic representation of the parafoil that requires only the vehicle position, heading and turning constraints. A high-fidelity aerodynamic model will be developed in future following CFD analysis and flight-based system identification.

7.5.4 Navigation

The Navigation module provides the Guidance subsystem with the vehicle state required for trajectory planning and control. The Navigation module does not perform state propagation; instead, it directly utilizes measurements provided by the GNSS receiver, barometer and ADCS.

The vehicle state is represented as

$$\mathbf{x} = \begin{bmatrix} N & E & h & \psi & V_N & V_E \end{bmatrix}^T, \tag{7.6}$$

where N and E denote the vehicle position in the local North-East-Down (NED) frame, h is the altitude above ground level, ψ is the vehicle heading, and V_N and V_E are the ground velocity components.

Vehicle position and ground velocity are obtained directly from the onboard GNSS receiver. The measured latitude and longitude are converted to the local NED frame, while the reported ground velocity components are used for trajectory planning. Altitude is obtained from the onboard barometer, providing the primary reference for parafoil deployment and terminal flare initiation.

The vehicle heading (ψ) and heading rate ($\dot{\psi}$) are obtained from the ADCS, which estimates the vehicle attitude using a Mahony filter. The resulting navigation state is continuously supplied to the Guidance module for trajectory generation and heading control.

7.5.5 Wind Estimation

Atmospheric wind significantly influences the ground trajectory of a parafoil. Since the CAN-7USAT does not employ dedicated air-data sensors, the horizontal wind components are estimated using the GNSS-derived ground speed together with the estimated trim airspeed.

The velocity relationship in the ground frame is given by

$$^{g}\mathbf{V}_g = {}^{g}\mathbf{V}_c + {}^{g}\mathbf{V}_w, \tag{7.7}$$

where $^{g}\mathbf{V}_g$, $^{g}\mathbf{V}_c$ and $^{g}\mathbf{V}_w$ denote the ground velocity, canopy airspeed and wind velocity expressed in the ground frame, respectively.

Assuming steady trimmed flight, the canopy airspeed is approximated by the calibrated trim airspeed, $\hat{V}_h$. The horizontal wind components are then estimated as

$$\hat{V}_{wh,x} = V_h \cos(\chi) - \hat{V}_h \cos(\psi), \tag{7.8}$$

$$\hat{V}_{wh,y} = V_h \sin(\chi) - \hat{V}_h \sin(\psi), \tag{7.9}$$

where V_h and χ are the horizontal ground speed and ground track angle obtained from the GNSS receiver, $\hat{V}_h$ is the estimated trim airspeed obtained through pre-flight calibration, and ψ is the vehicle heading obtained from the ADCS.

The trim airspeed is determined experimentally by averaging the steady-state ground speed measured during multiple straight, unbraked glide tests under low-wind conditions.

To reduce measurement noise, the estimated wind vector is filtered using an Exponential Moving Average (EMA),

$$\hat{\mathbf{V}}_{wh,k} = \alpha \hat{\mathbf{V}}_{wh,\text{measured}} + (1-\alpha)\hat{\mathbf{V}}_{wh,k-1}, \tag{7.10}$$

where α is the filter weighting factor. The filter is updated only during steady flight when

$$|\dot{\psi}| < \epsilon, \tag{7.11}$$

thereby avoiding corrupted estimates during turning manoeuvres. The filtered wind estimate is supplied to the trajectory planning module for wind compensation.

Figure 7.3: Wind estimation process used by the Guidance module.

7.5.6 Trajectory Planning

The trajectory planning module generates a dynamically feasible path from the current vehicle state to the designated landing target while satisfying the kinematic constraints of the parafoil. Since the deployment position and heading are unknown prior to launch, trajectory generation is initiated onboard after parafoil deployment and canopy stabilization using the estimated navigation state.

The parafoil is modelled as a forward-moving vehicle with a bounded turning radius. Under these constraints, the reference trajectory is generated using Dubins curves, providing the shortest continuously flyable path between the initial and target configurations. The generated trajectory is subsequently tracked by the Guidance module.

Figure 7.4 illustrates the trajectory generation process.

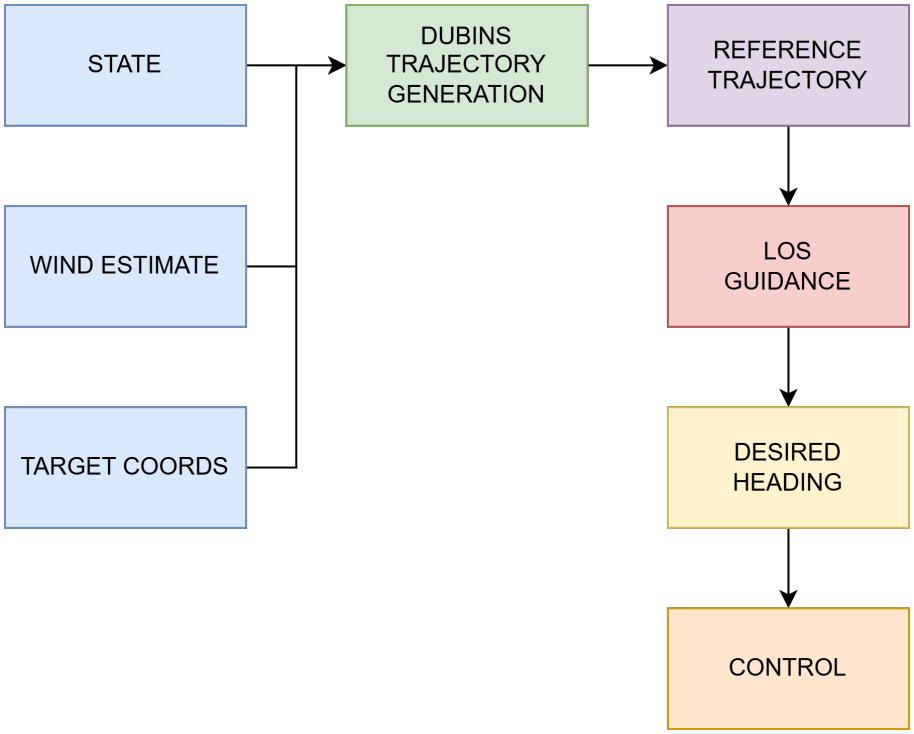


Figure 7.4: Overall trajectory planning process.

Initial Trajectory Generation

Once the parafoil has stabilized after deployment, the Navigation module provides the initial vehicle state,

$$\mathbf{x}_0 = \begin{bmatrix} N_0 & E_0 & h_0 & \psi_0 \end{bmatrix}^T, \tag{7.12}$$

from which the initial Dubins trajectory is generated. Since the initial conditions are only available after deployment, a new trajectory is computed for every mission.

Trajectory Replanning

During descent, the cross-track error between the vehicle and the reference trajectory is continuously monitored. If the deviation exceeds a predefined threshold, the existing trajectory is discarded and a new Dubins trajectory is generated from the current vehicle state.

Compared to optimization-based replanning methods, this geometric approach offers significantly lower computational complexity while remaining suitable for real-time execution on the STM32 flight computer.

7.5.7 Line-of-Sight Guidance

The generated trajectory is tracked using a Line-of-Sight (LOS) guidance law. Rather than steering directly towards the landing target, the guidance algorithm continuously computes a reference heading using a look-ahead point located on the reference trajectory. This results in smoother trajectory tracking while reducing heading oscillations and cross-track error.

The reference heading is computed as

$$\psi_{LOS} = \tan^{-1}\left(\frac{E_{LOS} - E}{N_{LOS} - N}\right), \tag{7.13}$$

where (N, E) and (N_{LOS}, E_{LOS}) denote the current vehicle position and look-ahead point, respectively.

The look-ahead distance L_d governs the responsiveness of the guidance algorithm. Smaller values improve tracking accuracy, whereas larger values provide smoother steering. During flight, the computed LOS heading is supplied directly to the heading controller.

7.6 Control

The objective of the Control module is to regulate the heading of the parafoil by converting the desired heading generated by the Guidance module into differential brake-line actuation.

The parafoil is controlled using two independent brake lines connected to servo actuators mounted within the CAN-7USAT. Differential deflection of these brake lines produces asymmetric aerodynamic drag, causing the parafoil to turn towards the desired heading. Simultaneous deflection of both brake lines is reserved exclusively for the terminal flare manoeuvre.

The controller operates continuously throughout the controlled descent and executes after each Guidance update. Figure 7.5 illustrates the overall control architecture.

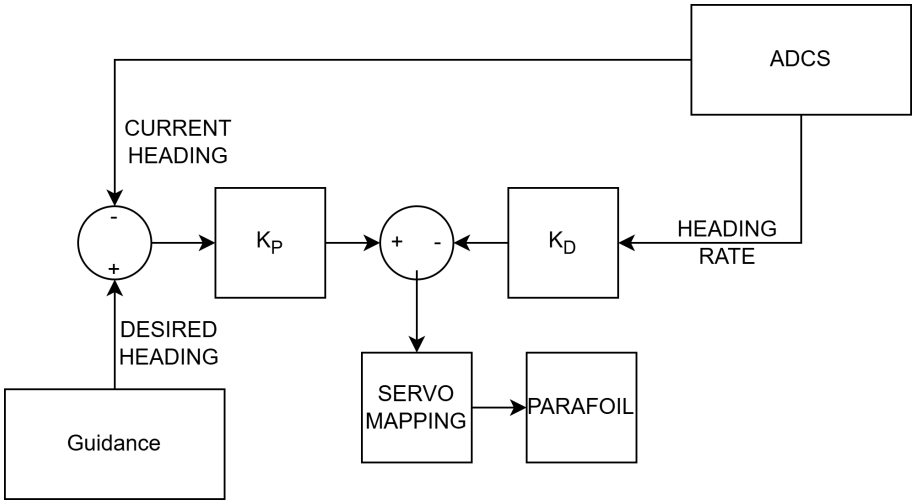


Figure 7.5: Heading control architecture.

7.6.1 Heading Error

The controller continuously compares the desired heading generated by the Guidance module with the measured vehicle heading obtained from the ADCS.

The heading error is defined as

$$e_\psi(t) = \psi_{desired}(t) - \psi(t). \tag{7.14}$$

To ensure the shortest rotational motion, the heading error is wrapped within the interval

$$-\pi \leq e_\psi \leq \pi. \tag{7.15}$$

This prevents discontinuities when the heading crosses the $\pm 180^\circ$ boundary.

7.6.2 PD Controller

A Proportional-Derivative (PD) controller is employed to regulate the vehicle heading.

The proportional term generates the steering command required to reduce the heading error, while the derivative term provides damping by opposing rapid rotational motion. This reduces oscillations arising from parafoil dynamics and improves overall heading stability.

The control effort is computed as

$$u(t) = K_p e_\psi(t) - K_d \dot{\psi}(t), \tag{7.16}$$

where

- K_p is the proportional gain,
- K_d is the derivative gain,
- $\dot{\psi}$ is the measured yaw rate obtained from the IMU.

The derivative term utilises the measured yaw rate directly rather than differentiating the heading error, thereby reducing amplification of measurement noise.

Controller gains will be determined experimentally through flight testing to achieve a satisfactory balance between response speed, tracking accuracy and oscillation damping.

7.6.3 Servo Mapping

The controller output is normalised within the interval

$$-1 \leq u \leq 1. \tag{7.17}$$

The steering command is converted into differential brake-line deflections according to the following logic:

Table 7.4: Servo Command Mapping

Control Output	Left Servo	Right Servo
$u > 0$	Neutral	Proportional Deflection
$u < 0$	Proportional Deflection	Neutral
$u = 0$	Neutral	Neutral

Only one brake line is actuated during normal flight. This minimises unnecessary drag and preserves the glide ratio while generating the required turning moment.

7.6.4 Servo Constraints

To avoid abrupt control inputs that may destabilise the canopy, the rate of servo motion is limited according to

$$\left|\frac{du}{dt}\right| \leq U_{max}. \tag{7.18}$$

Furthermore, the controller output is saturated to prevent excessive brake deflections that could induce canopy stall or significantly degrade glide performance.

These constraints improve flight stability while protecting the servo actuators from unnecessary mechanical loading.

7.6.5 Terminal Flare

During the final stage of descent, directional control is no longer prioritised. Once the vehicle descends below a predefined altitude threshold, the Guidance module transfers control to the landing routine.

Both brake lines are simultaneously actuated to produce a symmetric flare manoeuvre. Increasing the trailing-edge brake deflection momentarily increases the aerodynamic lift while reducing the forward velocity of the parafoil. This manoeuvre decreases the vertical impact velocity immediately before touchdown, resulting in a softer landing.

Following flare initiation, no further heading corrections are commanded until landing.

7.7 Components

The servo motors used for brake-line actuation were sized based on a conservative estimate of the maximum line tension. Assuming a maximum brake-line tension of 10 N, and considering that the parafoil load is shared across multiple suspension and control lines, the expected steady-state tension in an individual brake line is significantly lower. For a

two-brake-line configuration, the nominal tension in each brake line is expected to remain below 5 N.
Assuming a servo horn radius of 1 cm, the required actuation torque is

$$\tau = Fr = (5\,\mathrm{N})(1\,\mathrm{cm}) = 5\,\mathrm{N}\cdot\mathrm{cm}$$

To provide adequate margin against transient loads, gusts, and maneuvering conditions, a minimum torque capability of approximately 10 N·cm was selected.
The chosen actuator is the MG90S metal-gear servo, which offers a rated stall torque of approximately 17.65 N·cm while maintaining a compact form factor and low mass. The metal gear train also improves durability and resistance to wear under repeated actuation loads.

Chapter 8

Flight Software

8.1 Overview of Flight Software

The Flight Software (FSW) subsystem is responsible for the concurrent sensor polling loops, state machine transitions, telemetry collection and serialization. It processes guidance laws for stabilization and guided landing, update log tracking inside the local storage and command radio transmissions.

8.2 Changes Since PDR

Two architectural changes were made following the PDR.

8.2.1 Removal of RTOS

The PDR proposed an RTOS-based implementation. During further analysis, the identified flight-software workload was found to be compatible with a cooperative super-loop. Sensor acquisition, estimation, control, state transitions, communication, and logging do not require independent preemptive tasks when the slow peripheral operations are handled asynchronously.
The principal CPU-stalling operations were identified as:

- UART-based GNSS reception;
- XBee communication; and
- SD-card data transfers.

These interfaces are therefore implemented using DMA and interrupts. The remaining application functions are executed from the main loop according to software timing conditions.
This architecture removes task scheduling and inter-task synchronization overhead while retaining deterministic control over the execution order.

8.2.2 Removal of Pin-Based Mode Selection

The PDR proposed selecting simulation and flight operation using an external GPIO input. This approach would require test-specific behavior to be included in the flight executable. The CDR architecture removes the mode-selection pin. Testing is instead performed using dedicated test executables. This keeps the flight software independent of the test harness and prevents test-only logic from becoming part of the flight-state implementation.

8.3 Software Architecture

8.3.1 Architectural Layers

The software is divided into two principal layers.

Application Layer

The application layer consumes the data supplied by the drivers and performs mission-level processing. It is divided into the following functional modules:

Table 8.1: Application-layer functional responsibilities

Module	Responsibility
Sensor Acquisition	Obtain the latest valid sensor measurements and update the corresponding data structures.
State Machine	Determine the current mission state from sensor data and execute state-entry actions and state-dependent logic.
Attitude Estimation and Control	Estimate the current attitude using the available inertial and magnetic measurements. Compute reaction-wheel commands required for attitude stabilization.
DCS Guidance, Navigation and Control	Determine the desired descent trajectory and guidance commands during controlled descent. Convert guidance outputs into actuator commands for the descent-control servos.
Communication and Telemetry	Process ground commands, construct telemetry packets, transmit telemetry.
Logging	Construct the onboard telemetry record and transfer it to the SD card.

Driver Layer

The driver layer contains hardware-specific interfaces. Each driver encapsulates initialization, data reception/transmission, conversion where required, and access to the most recent valid data.
The principal drivers are for:

- AHT20+BMP280 pressure, humidity and temperature sensor;
- BMI270 inertial sensor;
- IIS2MDC magnetometer;
- NEO-6M GNSS receiver;
- XBee communication module;
- SD card reader interface;
- battery-voltage ADC;
- reaction-wheel ESC/PWM interface;
- servo PWM interface; and

- GPIO switches and indicators.

The driver interface shall expose application-independent functions such as `Init()`, `ReceiveData()`, `GetData()`, and actuator command functions. Hardware-specific register operations remain inside the driver.

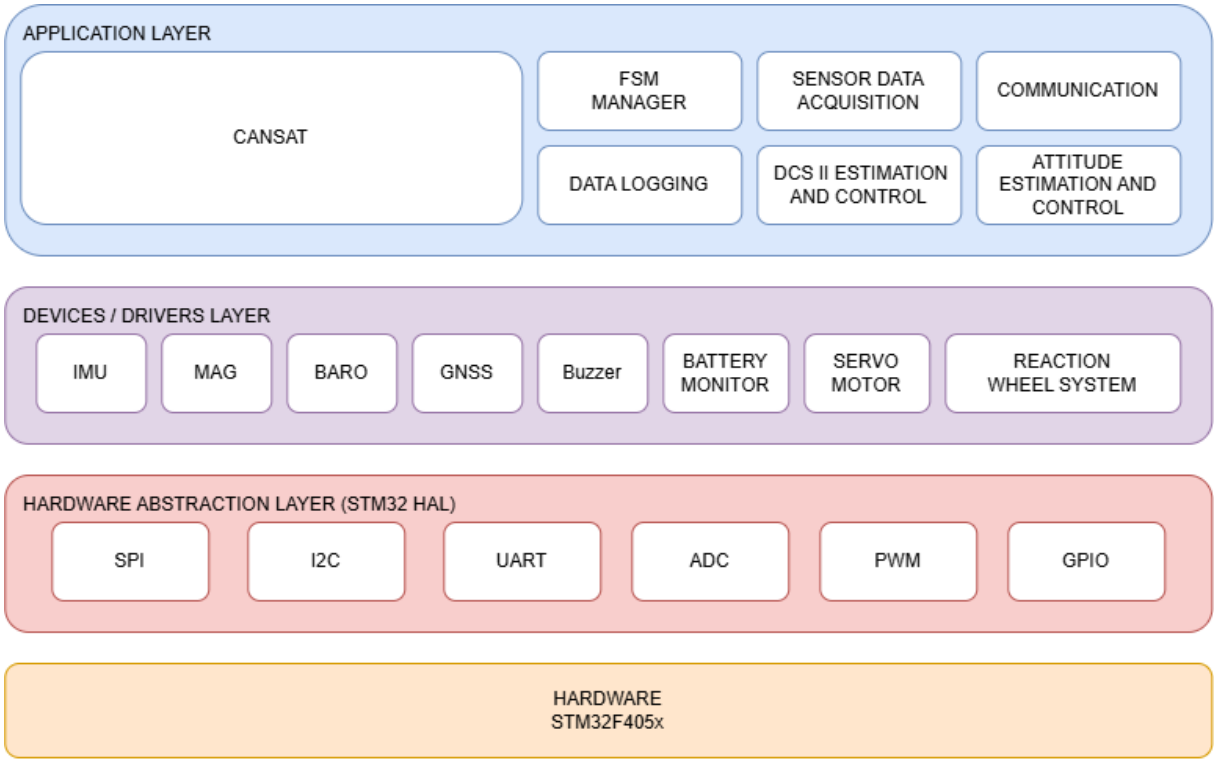


Figure 8.1: Software Architecture

8.3.2 State Machine

The software executes via a structural state machine configured to transition chronologically based on measured altitude. The state transition logics mentioned in the ConOps will be implemented in the software accordingly. The defined execution states are:

1. **BOOT:** Verifies system register configurations, initial peripheral checks, and opens interface files on the SD matrix
2. **TEST_MODE:** Executes debugging echo sequences and sensory calibration tests over temporary communication lines
3. **LAUNCH_PAD:** Samples steady ground pressure offsets to map relative flight altimetry scales to exactly zero
4. **ASCENT:** Detects continuous upward acceleration vectors and expanding altitude readings
5. **DESCENT_1:** Runs attitude estimation filtering upto 600 m attitude
6. **DESCENT_2:** Runs attitude estimation filtering, commands active feedback stabilization and parafoil guidance till touchdown
7. **TOUCHDOWN:** Switches of all sensors and process. Starts and plays the buzzer indefinitely.

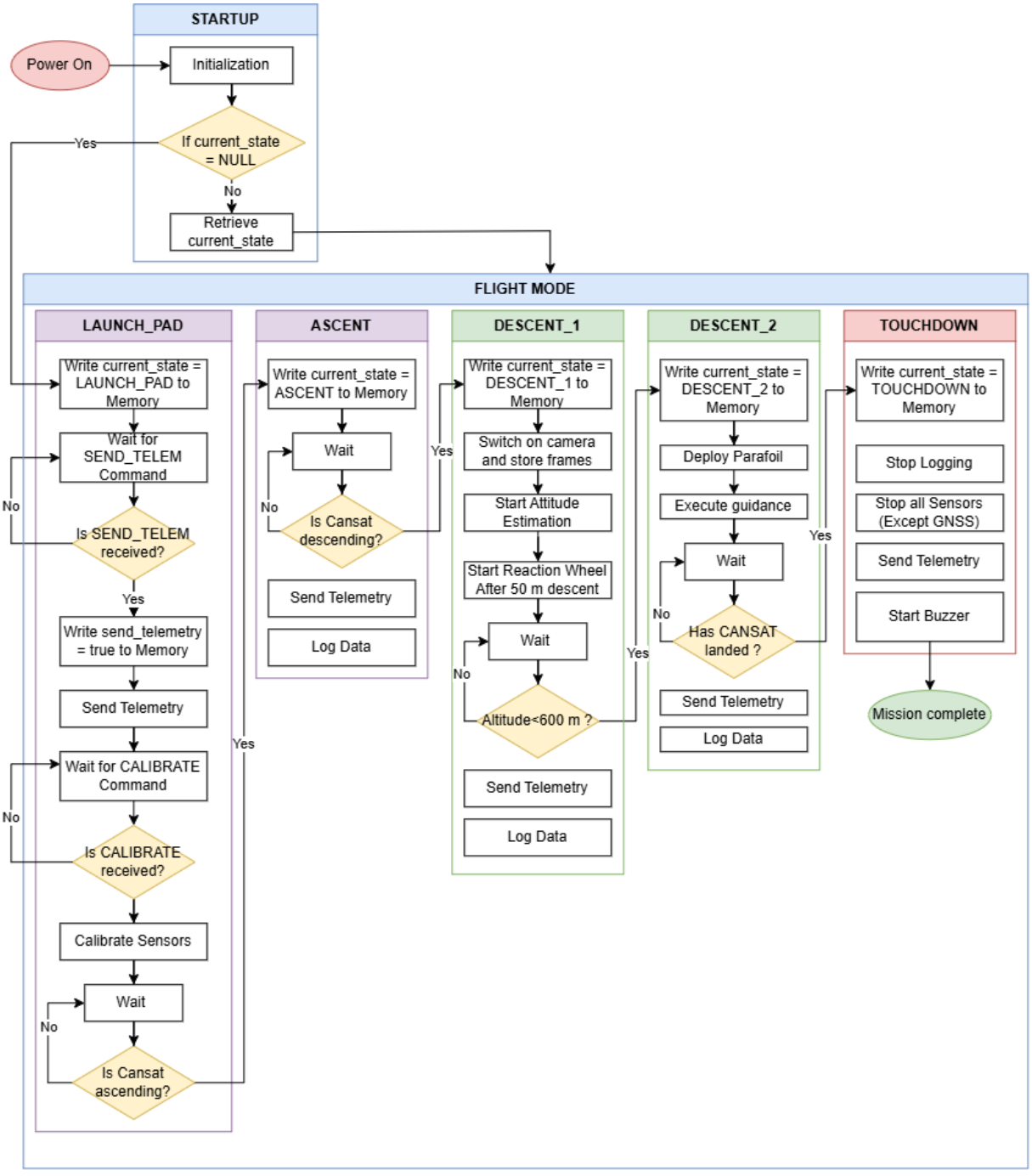


Figure 8.2: Operational Modes

8.4 Data Structures and Data Handling

8.4.1 Central Mission Data Structure

A central `Cansat` structure is used as the application-level data container. Component-specific structures hold the latest valid data from each hardware module, while the`Cansat` structure provides a common interface for application modules such as telemetry, logging, state determination, and control.
A representative organization is:

```
Cansat
|
+-- GNSSData
+-- BMPData
+-- BMIData
+-- MagnetometerData
+-- SwitchState
+-- BatteryData
+-- AttitudeState
+-- NavigationState
+-- ActuatorState
+-- MissionState
```

The exact field names and numerical types shall be defined later.

8.4.2 Memory Allocation

The data-storage strategy is divided according to persistence requirements.

Table 8.2: Flight-software data storage

Storage	**Data**	**Purpose**
Volatile RAM	Sensor structures, GNSS data, switch states, attitude and control data, central `Cansat` structure	Current mission data used by the flight loop.
Internal Flash	Mission state and firmware	State recovery across processor resets and persistent program storage.
SD Card	Telemetry records	Onboard backup of mission data and post-flight analysis.

Only the minimum state information required for recovery should be written to non-volatile memory during flight-state transitions. Continuous high-rate logging to internal flash is avoided to limit unnecessary flash writes.

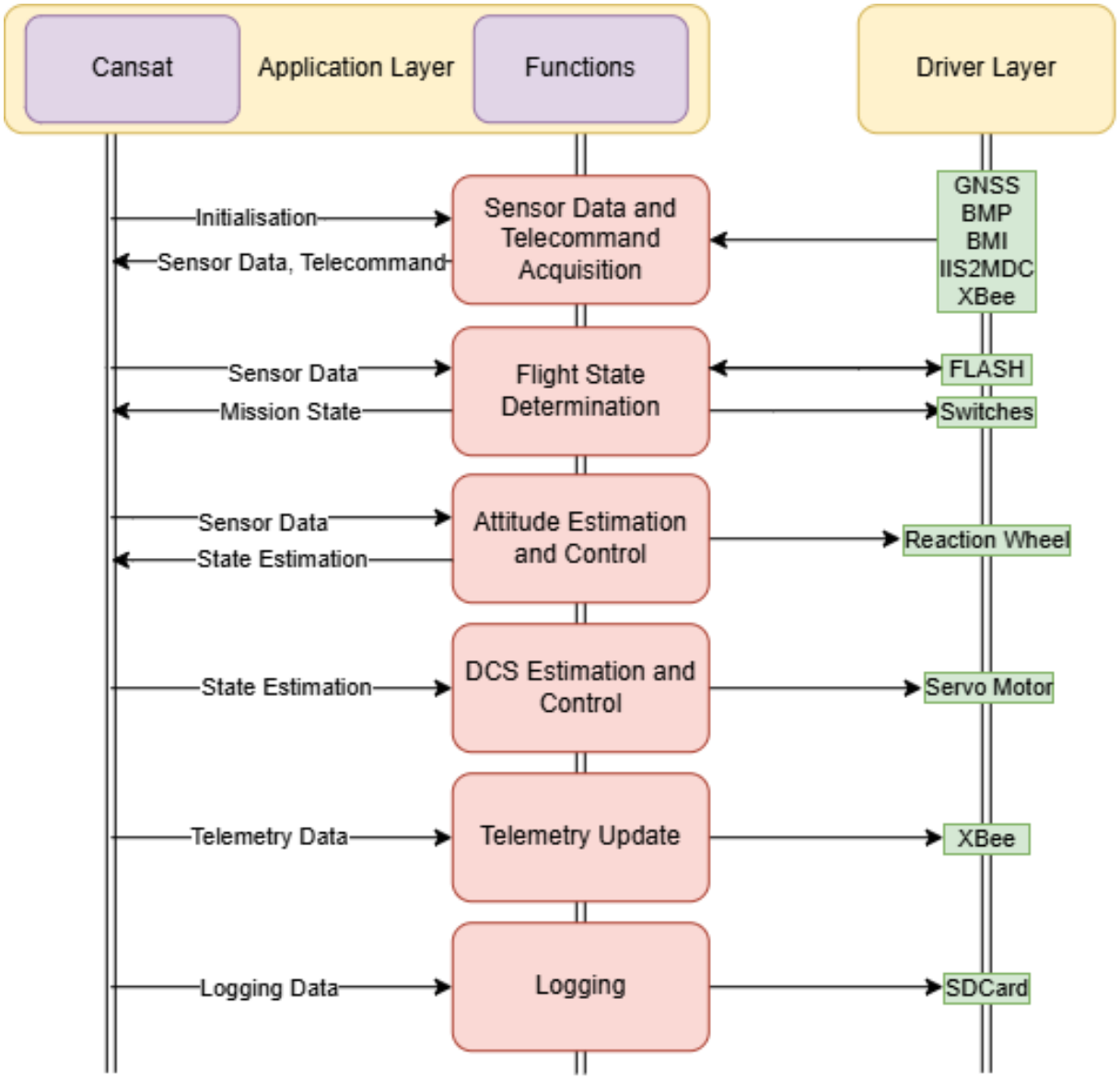


Figure 8.3: Enter Caption

8.5 Main Flight Loop Pseudocode

The flight software uses a cooperative super-loop. Each iteration checks the software timers associated with the application functions and executes a function when its timing condition is satisfied.
A conceptual implementation is:

```
while (1)
{
    if (sensor_timer_expired())
        SensorAcquisition_Update();

    if (fsm_timer_expired())
        StateMachine_Update();
```

```
    if (estimation_timer_expired())
        AttitudeEstimator_Update();

    if (control_timer_expired())
        Control_Update();

    if (telemetry_timer_expired())
        Telemetry_Request();

    if (logging_timer_expired())
        Logging_Request();
}
```

DMA completion callbacks are not used for executing complete application algorithms. Their primary role is to signal completion or availability of data and update the corresponding driver state. Processing can then be performed by the appropriate application function during the main loop.
This keeps interrupt service routines short and prevents large portions of mission logic from executing at interrupt level.

8.6 Execution Frequency and Timing

The section identifies relative execution frequency of the major software functions. The numerical values below represent relative priority/frequency, not final periods. Final execution periods shall be established through timing measurements and subsystem testing.

Table 8.3: Relative execution frequency of FSW functions

Function	**Relative frequency**	**Rationale**
Sensor Acquisition	3	Fast application-level acquisition; GNSS is asynchronous.
State Machine	2	State conditions need regular evaluation.
Attitude Estimation	3	Required at the control-processing rate.
Control	3	Requires regular execution during active control.
Telemetry	1	Competition requirement is 1Hz or better.
SD Logging	1	Logging rate need not match the control rate.

The execution frequency of a function shall not be interpreted as its hardware sensor update rate. For example, the GNSS receiver provides solutions at a much lower rate than

the MCU can execute its main loop. The flight software therefore reads the latest valid GNSS solution rather than waiting for a new solution on every loop iteration.
Similarly, telemetry and SD-card operations are initiated according to their required output rates, while the actual peripheral transfers proceed asynchronously.

8.7 Software Reset and Mission-State Recovery

The software is designed to recover from an MCU reset without assuming that the mission has restarted from the beginning. The persistent mission state is stored in internal flash during state transitions. The state stored in flash shall be updated at defined state transitions rather than continuously.
The reset sequence is:

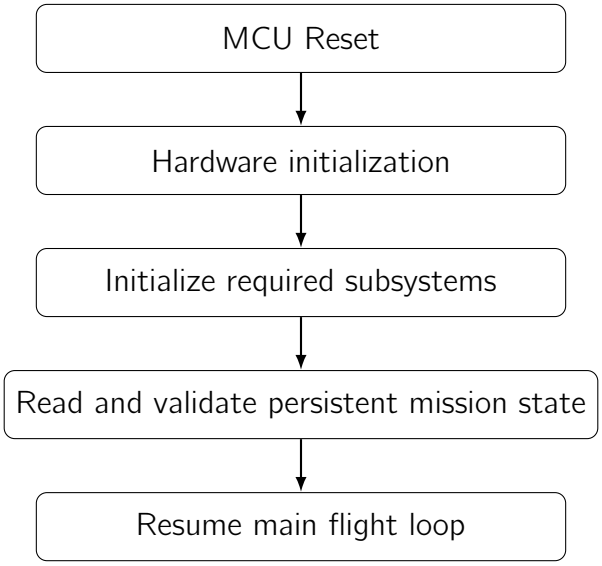


Figure 8.4: Software reset and mission-state recovery sequence.

8.8 Test Methodology

The test architecture separates flight software validation from the final flight executable. Individual tests are implemented as dedicated source files and are linked into separate executables. A test executable can access the required driver-layer or application-layer interfaces without modifying the production flight logic.
Two principal test categories are used.

8.8.1 Module Testing

Module tests validate individual hardware drivers and their interfaces. Tests are intended for components such as:

- AHT20+BMP280;
- BMI270;
- IIS2MDC;

- GNSS;
- XBee;
- SD card; and
- actuator interfaces.

The purpose is to verify initialization, communication, data conversion, error handling, and basic functionality before the module is used by the complete flight application.

8.8.2 Application and Simulation Testing

Simulation testing replaces the physical driver data source with an emulated data source. Input data can be supplied through a UART terminal and converted into the same application-level structures that would normally be populated by the hardware drivers.
The application therefore receives a controlled sequence of measurements without requiring physical flight conditions.
This method is intended to test:

- state-machine transitions;
- state-transition boundary conditions;
- recovery from processor reset;
- behavior under changing altitude and sensor inputs;
- telemetry generation;
- command processing; and
- interaction between estimation, guidance, and control modules.

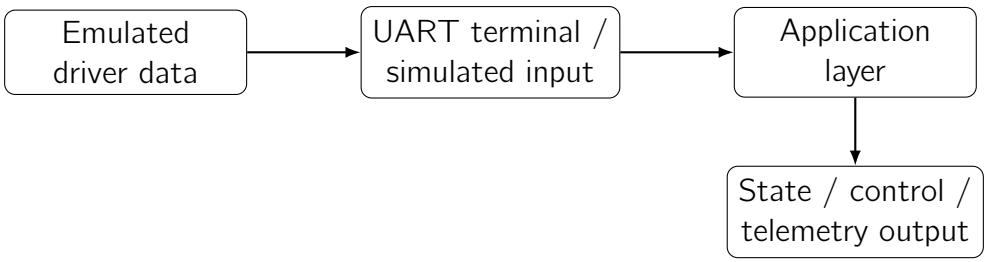


Figure 8.5: Application-level simulation test path.

8.9 Development Environment

The flight software is developed for the STM32 microcontroller platform. STM32CubeMX is used for peripheral configuration and code generation, while the project itself is maintained as a CMake-based embedded software project.
The generated low-level initialization and HAL components are integrated into the project build, while the application and driver source files are kept separate from generated code where practical.

The project is developed in C for the flight implementation, with hardware-facing code organized around explicit interfaces and structures.
Version control is used to track software changes. The CMake build system also allows the flight executable and dedicated test executables to be built from the same source tree.

Chapter 9

Ground Station Subsystem

9.1 Overview of Ground Station Subsystem

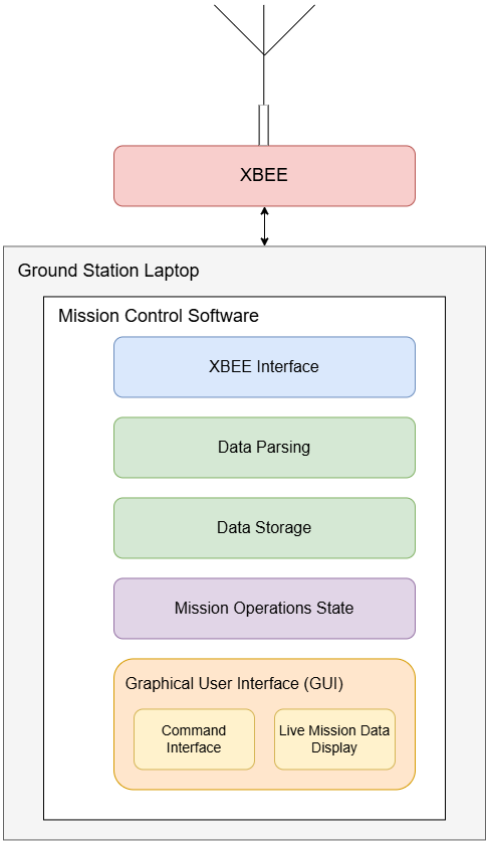


Figure 9.1: Ground Station

The Ground Station Subsystem is responsible for maintaining the RF communication link and performing the ground operations during the CAN-7USAT mission. The operations include commanding the CAN-7USAT over RF, receive and visually present the telemetry data transmitted by the CAN-7USAT and perform recovery operations.

9.2 Changes since PDR

- Changed antenna holding from hand-held to tripod mounted

9.3 Components



(a) Ground Station Yagi Antenna (b) Tripod

Figure 9.2: Ground Station Components

The ground station consists of an **XBEE PRO S2C** RF module connected to a directional high-gain **Yagi-Uda antenna** via a low-loss RF coaxial cable. The antenna is mounted on a tripod to elevate it from the ground. The ground station laptop is interfaced with the XBEE radio and runs the mission control software which is responsible for logging of mission operations data and presenting it over a graphical user interface (GUI). The ground station laptop is battery powered and is capable of handling more than 2 hours of operation.

Parameter	Value
No. of Elements	14
Frequency Range	2400 - 2483 MHz
Nominal Impedance	50 Ω
VSWR	$\leqslant$ 1.5:1
Gain	14 dBi
F/B Ratio	28 dBi
Polarisation	Linear
Horizontal HPBW	35°
Vertical HPBW	36°
Material	Aluminum

Table 9.1: Yagi Antenna Specifications

9.4 Mission Operations

The ground station performs mission operations as identified in the concept of operations. These operations include:

- Maintain mission logs
- Commission CAN-7USAT Communication Subsystem

- Performing ground checks before launch vehicle integration
- Command CAN-7USAT to calibrate onboard sensors
- Track the CAN-7USAT during flight
- Recover the CAN-7USAT post touchdown
- Safely disable the CAN-7USAT post recovery
- Retreive onboard data storage
- Post flight mission data analysis

9.5 Mission Control Software (MCS) and Graphical User Interface (UI)

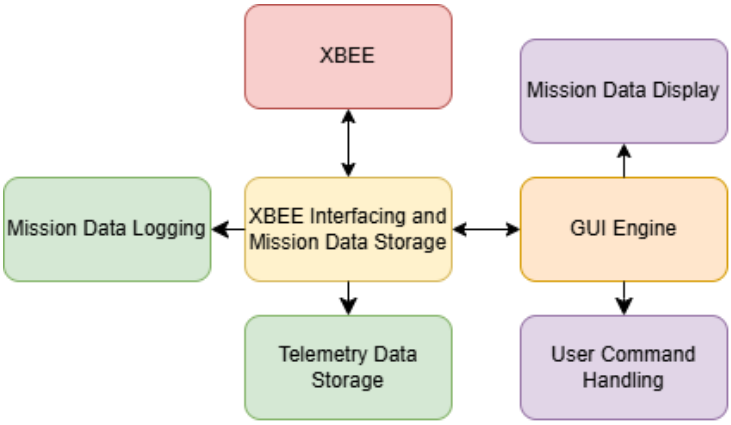


Figure 9.3: Mission Control Software Architecture

The MCS is a Python script running on the ground station laptop as a multithread process, making use of the following libraries:

Library	Function
PySerial	UART interfacing with XBEE
PySide6	GUI elements, multithreading
PyQtGraph	Real-time graph plotting

Table 9.2: Libraries being used in MCS



Figure 9.4: MCS GUI with dummy values

The first thread is responsible for maintaining GUI on the ground station laptop, while the other interfaces with the XBEE and maintains local data storage of mission data.
The data received over the radio is parsed from its ASCII comma separated format and saved in a .csv format file with the headers as specified in the table 5.4.
The MCS also logs the important mission events including mission operations states change, CAN-7USAT boot-up, transmission of telecommands and reception of ACK.
The GUI of the MCS contains real-time tracking panels for payload and various sensors data, visual indicator reflecting the current state machine of the CAN-7USAT, display of the last sent command and dedicated features for sending sensor calibration and telemetry toggle commands down to the launch pad.

9.5.1 Redundancy Verification

If a temporary signal loss occurs, the ground station software can append the missing data in visually distinguished manner on the GUI from the onboard SD card storage data retrieved post recovery.

Chapter 10

CAN-7USAT Integration and Testing

This section describes the methodology and test plans used to integrate all subsystems of the CAN-7USAT and verify that the integrated system meets the mission requirements of IN-SPACe CAN-7USAT India 2026. All tests shall be documented with pass/fail criteria, date of execution, and any deviations observed.

10.1 Integration Approach

Integration follows a bottom-up approach: each subsystem is independently verified before being integrated with adjacent subsystems and finally evaluated as a complete unit. The integration sequence is as follows:

Step 1. Electrical Subsystem (EPS) bench test

Step 2. Flight Software bring-up; sensor interfacing and calibration.

Step 3. Communication subsystem link test

Step 4. Camera interfacing and DVR checks.

Step 5. Descent control mechanism (primary parachute & secondary mechanism) functional tests.

Step 6. Full CAN-7USAT assembly with housing fit-check

Step 7. End-to-end functional test (TBD)

10.2 Subsystem Test Plans

10.2.1 Electrical subsystem

Table 10.1: Electrical Subsystem Test plan

#	Test Name	Description / Pass Criterion	Equipment
1	GNSS Lock Test	Acquire $\geq$4 satellites outdoors; verify latitude, longitude, altitude output at required resolution	GNSS receiver
2	Barometric Altitude Calibration	Verify 0.1 m resolution; compare against known elevation	BME280
3	Temperature Sensor Accuracy	Measure ambient temperature; compare with calibrated reference thermometer; tolerance $\pm$0.5 °C	BME280, Calibrated thermometer
4	IMU / Gyro Calibration	Zero gyro bias post calibration, characterise drift; verify roll, pitch, yaw output in deg/s	BMI270
5	Accelerometer Shock Test	Drop CAN-7USAT from 0.5 m; verify no data dropout, accelerometer reads $>$9.8 m/s^2 at impact	BMI270
6	Magnetometer calibration	Minimal scaling and offset post calibration; IIS2MDC	
7	SD Card Data Logging	Verify all sensor data written to SD card without corruption	SD card
8	Voltage regulator functional test	Verify regulated output of 3V3 & 5V for all nominal modes of voltage regulators	LTC3624-3.3, LTC3624-25
9	Current limiter test	Verify current limit function at regulated power rails of CAN-7USAT	TPS25200, TPS25926
10	Battery Capacity Test	Operate all systems for $\geq$3 h; bus voltage stays above minimum operating level	CAN-7USAT
11	External Switch Test	Toggle external power switch; indicator LED/buzzer activates; no system disassembly required	CAN-7USAT
12	Battery Swap Test	Remove and reinstall battery in $<$60 s without total disassembly	Stopwatch
13	Power Budget Verification	Measure current draw of each subsystem; confirm total $\leq$ budget	Bench power supply

10.2.2 Mechanical Subsystem Tests

Table 10.2: Mechanical Subsystem Test Plan

#	Test Name	Description / Pass Criterion	Equipment
1	Housing Fit-Check	Verify CAN-7USAT fits within 0.15 m dia × 0.40 m height cylinder; no protrusions; no sharp edges	Tape measure for course measurement
2	Mass Verification	Total mass $\leq$1000 g (1000 $\pm$ 50 g)	Precision scale ($\pm$1 g)
3	Vibration Test	Subject to vibration profile simulating 15 G launch & 30g shock; no structural failure or connector loosening	Shake-test rig
4	Thermal Test	Operate at $-5\,^{\circ}$C to $+60\,^{\circ}$C for 30 min each; verify normal operation	Thermal Chamber

10.2.3 Communication and Data Handling (CDH) Tests

Table 10.3: Communication Subsystem Test Plan

#	Test Name	Description / Pass Criterion	Equipment
1	Link Range Test	Achieve reliable telemetry link at $\geq$1 km LOS	XBEE
2	RF Interference Test	No interference with adjacent radios	Multiple radio set
3	Telemetry Test	Verify ASCII CSV packet format matches specification	Ground Station
4	RF Command Test	Ground station sends ON/OFF telemetry command via RF; CAN-7USAT send ACK	CAN-7USAT; Ground Station
5	Gyro Zero Command	Ground station issues calibration command; verify parameters reset while CAN-7USAT is stationary	CAN-7USAT; Ground station

10.2.4 Flight Software Tests

Table 10.4: Flight Software test plan

#	Test Name	Description / Pass Criterion	Equipment
1	Processor Reset Recovery	Simulate processor reset mid-mission; verify mission clock/timestamp and system state are preserved	MCU
2	State Machine Test	Check all operation and contingency modes	MCU

10.2.5 Attitude Determination and Control Test plan

Table 10.5: Attitude Determination and Control System test plan

#	Test Name	Description / Pass Criterion	Equipment
1	Yaw Stabilisation Test	Power on reaction wheel; verify CAN-7USAT yaw stabilizes in free-spin	Free-hanging rig
2	Attitude Estimator test	Turn on attitude estimator, IMU and magnetometer; Verify orientation accuracy	BMI270, IIS2MDC, MCU

10.2.6 Ground Station Tests

Table 10.6: Ground Station Test Plan

#	Test Name	Description / Pass Criterion	Equipment
1	Real-Time Display Test	All telemetry fields displayed in SI units; plots update at $\geq$1 Hz	Ground Station
2	CSV File Generation	After 5-min simulated flight, verify `Flight_TEAMID.csv` generated with correct headers; transferable via USB	Ground station
3	Battery Endurance	GS laptop operates on battery for $\geq$2 h	Ground station laptop

10.2.7 Parachute and Descent Control Tests

Table 10.7: Descent Control System Test Plan

#	Test Name	Description / Pass Criterion	Equipment
1	Primary Parachute Deployment	Deploy parachute immediately after ejection (manual toss);	Stopwatch
2	Parafoil Deployment	Test secondary DCS burn-wire deployment and canopy inflation at a specified altitude ($\pm$10 m);	Dummy CAN-7USAT, BME280
3	Parafoil Guidance & Control	Verify steering control, log data, verify dynamics model and approach performance	Parafoil deployment mechanism, MCU

10.3 Integrated Level Functional Test

Upon completion of all subsystem tests, a full end-to-end functional test shall be conducted with the complete assembled CAN-7USAT and the ground station. The test shall simulate the complete mission sequence, exercising each flight software state.
All telemetry shall be logged to a CSV file and reviewed for completeness and correctness against the specifications defined in table 5.4 of the mission requirements.

Chapter 11

Budget

This section presents the complete Bill of Materials (BOM) and cost breakdown for the CAN-7USAT project. The maximum permissible cost is **Rs. 1,00,000/-** (Rupees One Lakh Only), inclusive of material, component, and facility usage costs. Priority has been given to *Made-in-India* components wherever available.

11.1 Bill of Materials (BOM)

Due to the length of the BOM, it is presented across category-wise tables below.

Table 11.1: BOM – A. Mechanical Structure

#	Component	Material/Model	Qty	Unit (Rs.)	Est. (Rs.)
1	Cylindrical body / airframe	ABS	1 set	1100	1100
2	PCB standoffs & screws	M3 standoffs	1 set	30	30
3	Parachute	Custom fabric	1	2000	2000
4	Parafoil	Custom fabric	1	5000	2000
5	Secondary descent mechanism	Servo motors	1	500	500
6	Mechanical reaction wheel	BLDC Motor	1 set	2000	2000
7	Audio beacon ($\geq$92 dB)	AI-4228-TWT-R	1	525	525
8	Fasteners, adhesives, misc.	Various	–		500
				Sub-total A	8655

Table 11.2: BOM – B. Electronics and On-Board Computer

#	Component	Make/Model	Qty	Unit (Rs.)	Est. (Rs.)
9	Microcontroller	STM32F405RG	1	1267	1267
10	GNSS / NavIC receiver	PX1125s-01A	1	2476	2476
11	Pressure and temp. sensor	BME280	1	530	530
12	IMU / Gyro sensor	BI270	1	466	466
13	DVR	RunCam Mini FPV	1	2200	2200
14	PCBs (custom)	<Fab>	1	5000	5000
15	Servo	MG90S	2	119	238
16	BLDC Motor + ESC	A2212 1000 kV	1	1710	1710
17	Camera	FOXEER PREDATOR 5 Nano FPV	1	5000	5000
18	Magnetometer	IIS2MDC	1	320	320
				Sub-total B	19207

Table 11.3: BOM – C. Communication

#	Component	Material/Model	Qty	Unit (Rs.)	Est. (Rs.)
19	Communication Module (onboard)	XBEE Pro S2C	2	15000	30000
20	Onboard Dipole Antenna	ANT-2.4-CW-HWR-RPS	1	2700	2700
21	GS Yagi-Uda antenna	YAGI-14-2.4G	1	5836	5836
				Sub-total C	38536

Table 11.4: BOM – D. Electrical Power E. Design & Fabrication

#	Component	Material/Model	Qty	Unit (Rs.)	Est. (Rs.)
D. Electrical Power					
22	Li-ion 18650 cells	`Samsung INR18650`	3	392	1176
23	Voltage regulator/ current limiter	`<IC>`	1	1673	1673
24	External power switch + indicator LED	–	1	50	50
E. Design, Fabrication & Testing					
25	3D printing / machining	–	–	10000	10000
F. Contingencies & Miscellaneous					
26	Contingency reserve (∼10%)	–	–	6910.2	6910.2
			Sub-total D+E+F		19809.2

11.2 Cost Summary

Table 11.5: Project Cost Summary

Category	Estimated (Rs.)	Actual (Rs.)
A. Mechanical Structure	8655	
B. Electronics and OBC	19207	
C. Communication	38536	
D. Electrical Power	2899	
E. Design, Fabrication & Testing	10000	
F. Contingencies & Misc.	6910.2	
TOTAL	**86207.2**	
Maximum Allowed	**Rs. 1,00,000/- (One Lakh Only)**	

Remarks:

- Registration fee (Rs. 10,000/-) and CDR-stage fee (Rs. 15,000/-) are paid separately per competition guidelines and are not included in the above component cost.

Chapter 12

Development Schedule

This section presents the project milestone chart showing task start and end dates and durations for the complete CAN-7USAT development lifecycle. The schedule aligns with the IN-SPACe competition timeline and includes internal milestones for design, procurement, fabrication, testing, and review submissions. Periodic progress-review meetings are scheduled every two weeks.

12.1 Competition Timeline Reference

Table 12.1: IN-SPACe CAN-7USAT India 2026 Official Competition Timeline

Phase	Activity	Start Date	End Date
1	Registration + fee payment	24-03-2026	30-04-2026
2	Evaluation and screening	01-05-2026	15-05-2026
3	PDR Document Submission	16-05-2026	15-06-2026
4	PDR completion by jury	01-06-2026	30-06-2026
5	CDR document submission + fee	01-07-2026	31-07-2026
6	Comprehensive Design Review completion	01-08-2026	31-08-2026
7	Flight Readiness Review (FRR)	01-09-2026	15-10-2026
8	Launch window, Post-Flight Review, Results	16-10-2026	30-11-2026

12.2 Team Development Gantt Chart

The following Gantt chart presents the internal project development schedule. Each unit on the horizontal axis represents one fortnight. The **red dashed line** marks the current date. Dates are to be updated as the project progresses.

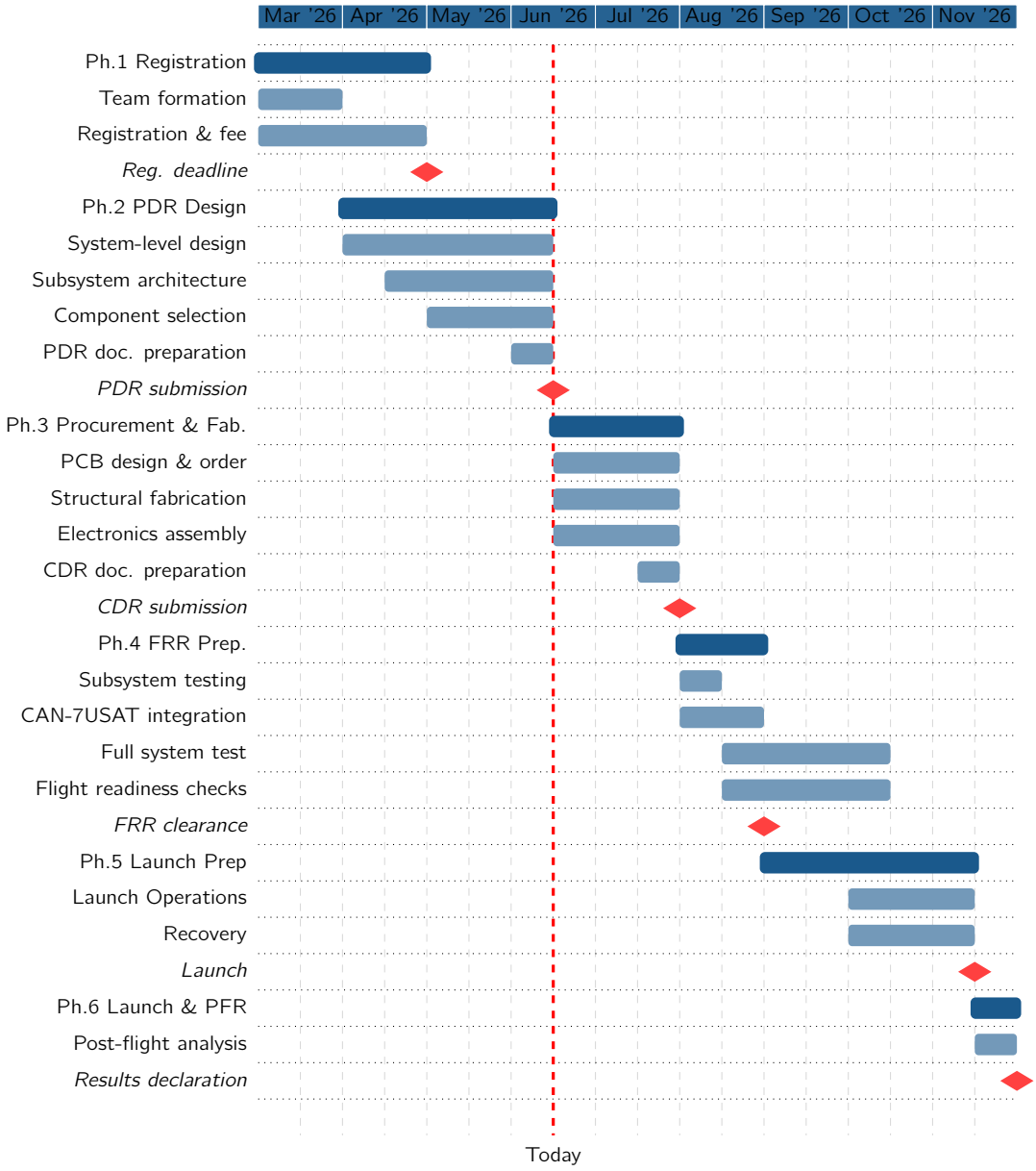


Figure 12.1: CAN-7USAT Development Schedule – Gantt Chart (fortnightly resolution)

12.3 Detailed Milestone Table

Table 12.2: Detailed Project Milestone Table – Phases 1 to 3

#	Milestone / Task	Planned Start	Planned End	Responsible
Phase 1 – Registration & Concept Design				
1	Team formation and role assignment	Mar 2026	Mar 2026	Team Leader
2	IN-SPACe registration and fee payment	Mar 2026	Apr 2026	Faculty Advisor
3	Mission requirement review and concept selection	Mar 2026	Apr 2026	All
Phase 2 – PDR Design				
4	System-level architecture finalisation	Apr 2026	May 2026	Systems Eng.
5	Subsystem design (Mech., EPS, COMM, sensors)	Apr 2026	May 2026	Sub-teams
6	Component selection and vendor identification	May 2026	May 2026	All
7	Simulation and preliminary analysis	Apr 2026	May 2026	Structures, COMM, EPS
8	PDR document writing and internal review	May 2026	May 2026	All
9	**PDR document submission**	**15-06-2026**	**15-06-2026**	Team Leader
10	PDR presentation to jury	Jun 2026	Jun 2026	Team Leader
Phase 3 – Procurement & Fabrication				
11	Component procurement	Jun 2026	Jul 2026	Procurement
12	PCB design and fabrication order	Jun 2026	Jul 2026	Electronics
13	Structural fabrication (body, mounts, parachute)	Jun 2026	Jul 2026	Mech.
14	Electronics assembly and bring-up	Jul 2026	Jul 2026	Electronics
15	CDR document preparation	Jul 2026	Aug 2026	All
16	**CDR document submission**	**01-07-2026**	**31-07-2026**	Team Leader
17	CDR presentation and jury response	Aug 2026	Aug 2026	Team Leader

Table 12.3: Detailed Project Milestone Table – Phases 4 to 6

#	Milestone / Task	Planned Start	Planned End	Responsible
Phase 4 – Subsystem Testing & CDR				
18	Subsystem-level testing (see Section 10)	Jul 2026	Aug 2026	All
19	CAN-7USAT integration	Jul 2026	Aug 2026	Systems Eng.
20	Integrated end-to-end functional test	Aug 2026	Aug 2026	All
Phase 5 – Flight Readiness Review				
21	Final system-level test and FRR checks	Sep 2026	Oct 2026	All
22	Pre-flight requirement compliance check	Sep 2026	Oct 2026	Systems Eng.
23	**FRR clearance**	**15-10-2026**	**15-10-2026**	IN-SPACe jury
Phase 6 – Launch & Post-Flight Review				
24	Arrival at launch site and pre-launch checks	Oct 2026	Nov 2026	All
25	CAN-7USAT launch	Oct 2026	Nov 2026	Launch agency
26	Post-flight data analysis and report	Nov 2026	Nov 2026	All
27	**Results declaration**	**30-11-2026**	**30-11-2026**	IN-SPACe

12.4 Review Meeting Schedule

Periodic progress-review meetings shall be held every two weeks throughout the competition. Minutes shall be recorded and shared with the faculty advisor and mentor.

Table 12.4: Internal Review Meeting Schedule

#	Review Topic	Target Date	Expected Outcome
1	Kickoff + mission requirements	30 Apr 2026	Role assignments confirmed
2	System architecture review	15 May 2026	Architecture baselined
3	PDR internal review	30 May 2026	PDR document ready for submission
4	Component procurement status	30 Jun 2026	All parts ordered
5	Fabrication progress review	1 Jul 2026	Hardware $\geq$50% complete
6	Subsystem test results review	15 Jul 2026	All subsystem tests passed
7	CDR internal review	20 Jul 2026	CDR document ready
8	Integration test review	15 Aug 2026	Full system functional
9	FRR preparation review	1 Sep 2026	Flight checklist complete
10	Pre-launch briefing	10 Oct 2026	Launch readiness confirmed

Appendix A

Team Details

Team Name: Samarthya
Team ID: 2026-IN-SPACe-CAN-7USAT-016

Details of the team is tabulated below:-

Sr.	Name	Program	Graduation Year	Department	Role
1	Dr. Rohit Nanavati	-	-	-	Faculty Advisor
2	Manyaman Naik	BTech	2027	Mechanical Engineering	Team Lead
3	Pratyush Masanta	BTech	2028	Electrical Engineering	Team Member
4	Aditya Nimbalkar	BTech	2027	Mechanical Engineering	Team Member
5	Yugratna Shaurya	Dual Degree (BTech + MTech)	2029	Electrical Engineering	Team Member
6	Anant Kumar Singh	Dual Degree (BTech + MTech)	2029	Electrical Engineering	Team Member
7	Naigam Sanghvi	BS	2027	Center for Multidisciplinary Education	Team Member
8	Shishir V Shetty	BTech	2029	Physics	Team Member